# Path-Integral Tolerance Analysis in Optical Engineering: A Fast Alternative to Monte Carlo

From Fabrication Disorder through Thermal Drift to Quantum Yield Advantage

**Version History:**
- **v0 (Mar 4, 2026):** Early LinkedIn post: "On My Upcoming Physics Colloquium Talk at NCU (March 4, 2026)"; thoughts on worked examples from basic to advanced
- **v7 (Mar 20, 2026):** Preliminary technical document.

**Abstract**

Tolerance and yield analysis is the most time-consuming bottleneck in photonic device development. The challenge intensifies for integrated photonic circuits, where fabrication disorder, thermal drift, and mechanical vibration interact across dozens of components on a single chip; it becomes acute for quantum photonic devices, where entanglement fidelity imposes specifications far tighter than classical telecom.

This document presents the **path-integral (PI) approach to tolerance analysis** as a structurally efficient alternative to Monte Carlo (MC) simulation. For Gaussian fabrication disorder — the dominant model for sidewall roughness and thickness variation in silicon photonics — the PI computes the mean and variance of the propagator *analytically in a single calculation*, achieving speedups of $10^2$–$10^3\times$ over MC for standard components, $10^8\times$ for chaotic cavities (Gutzwiller trace formula), and infinite speedup where the Strehl–fidelity identity $S = F$ eliminates a separate quantum computation entirely.

The framework is demonstrated through **eight worked examples** spanning three tiers of device complexity (TRL 2–9): directional couplers, microring resonators, MMI couplers, Mach–Zehnder interferometers, adiabatic tapers, photonic crystal cavities, deformed microdisk cavities, and Möbius metasurfaces. For each case, the self-energy tolerance analysis is computed, classical and quantum yields are compared, and design rules are extracted.

Three results appear to be new. First, the PI reveals a **quantum yield advantage**: entangled photon pairs are partially immune to fabrication disorder through parity cancellation in the two-particle path integral, giving $Y_{\mathrm{qu}} \geq Y_{\mathrm{cl}}$ with the advantage growing linearly with system complexity. Second, the **Maslov phase budget** ($\Delta\phi = -\pi\mu/2$ per cutoff crossing) identifies a systematic phase error in PIC routing that is invisible to all commercial EDA tools. Third, the **influence functional** predicts wider tolerance windows than the Lindblad equation for structured environments, because non-Markovian entanglement revival extends device operation beyond the Markovian prediction.

Four bridge equations are formulated as operational switching points, each with an explicit monitoring observable, alarm threshold, and corrective action, transforming the PI framework from abstract theory into actionable engineering practice. A literature survey across all eight device classes identifies where prior tolerance specifications existed and where the PI framework fills gaps — most notably for the Maslov phase (no prior specification) and topological retardance tolerance (no prior quantification).

This preliminary technical document aims to bridge classical and quantum optical engineering, enabling the path-integral tolerance framework as a practical design tool for the emerging quantum photonic foundry ecosystem.

## Contents

**List of Abbreviations** **4**

**1 The Pain Point: Why Tolerance Analysis Is Hard** **5**
1.1 The Production Engineer's Dilemma . . . 5
1.2 Tolerance Analysis in the Early Literature: What Was Known . . . 5
1.3 What the Path Integral Offers . . . 7
1.4 Scope and Honest Assessment . . . 8

**2 Three Classes of Tolerance: Physics, Failure Modes, and Why PI Helps** **8**
2.1 Class A: Fabrication Disorder . . . 8
2.2 Class B: Thermal Perturbation . . . 9
2.3 Class C: Mechanical and Environmental Perturbation . . . 9
2.4 Summary: Good and Bad of Each Approach . . . 10

**3 Analytical Procedures** **11**
3.1 Fabrication Tolerance (Class A) . . . 11
3.2 Thermal Tolerance (Class B) . . . 11
3.3 Mechanical Tolerance (Class C) . . . 11
3.4 Combined Yield Model . . . 11

**4 The Quantum Yield Advantage: Why Entanglement Helps Production** **11**
4.1 Physical Intuition: Why Should Entanglement Help? . . . 12
4.2 Derivation Sketch: From Two-Particle Path Integral to $S_{12} > S^2$ . . . 12
4.3 When the Advantage Is Largest . . . 13
4.4 System-Level Scaling: Why $\Delta Y$ Grows with Complexity . . . 14
4.5 Practical Design Guidance: How to Utilize the Advantage . . . 14
4.6 When the Advantage Does Not Help . . . 15

**5 Eight Worked Examples** **15**
5.1 Case 1: Directional Coupler (SOI) . . . 15
5.2 Case 2: Microring Resonator (SOI/SiN) . . . 16
5.3 Case 3: MMI Coupler for Boson Sampling (SOI) . . . 17
5.4 Case 4: Mach–Zehnder Interferometer (SOI/$LiNbO_3$) . . . 18
5.5 Case 5: Adiabatic Taper / Mode Converter (SOI) . . . 19
5.6 Case 6: Photonic Crystal Cavity (Si/InP) . . . 20
5.7 Case 7: Deformed Microdisk / Stadium Cavity (InP) . . . 21
5.8 Case 8: Möbius Metasurface (Topological) . . . 22

**6 Cross-Case Comparison: Lessons Learned and Mixed-Case Guidance** **24**
6.1 Lessons Learned from the Eight Cases . . . 24
6.2 Mixed-Case Systems: When Devices Combine on One Chip . . . 25
6.2.1 How Tolerances Compound in Mixed Systems . . . 26
6.2.2 Priority Hierarchy: What to Fix First . . . 26
6.2.3 Worked Guidance: A Realistic Quantum PIC . . . 27
6.2.4 Warning: What Can Go Wrong in Mixed Systems . . . 28

**7 Bridge Equations: Switching Points, Control, and Alarm Mechanisms** **28**
7.1 Bridge #2: $S = F$ (Classical Strehl = Quantum Fidelity) . . . 29
7.2 Bridge #4: Gutzwiller ($\rho(E)$ from Periodic Orbits) . . . 29
7.3 Bridge #5: Feynman–Vernon ($\gamma_k = \pi J(\omega_k)$) . . . 30
7.4 Bridge #6: Maslov ($\Delta\phi = -\pi\mu/2$) . . . 30
7.5 Bridge Alarm Summary . . . 31

**8 Implementation: Workflow and Reporting** **31**
8.1 Companion Code . . . . . 31
8.2 Quality Checklist . . . . . 32

**9 The Overarching Finding** **32**

# List of Abbreviations

***Formalisms and Methods***

ABCD – Ray-transfer matrix
BPM – Beam propagation method
CMT – Coupled-mode theory
DFT – Discrete Fourier transform
FDTD – Finite-difference time-domain
GOE – Gaussian orthogonal ensemble
IF – Influence functional
MC – Monte Carlo
PI – Path integral
PSD – Power spectral density
RSS – Root-sum-square
WKB – Wentzel–Kramers–Brillouin

***Quantum Optics and Information***

ESD – Entanglement sudden death
HOM – Hong–Ou–Mandel effect
JC – Jaynes–Cummings model
LOQC – Linear optical quantum computing
NOON – $N$-photon path-entangled state
QKD – Quantum key distribution
SFWM – Spontaneous four-wave mixing

***Device Types and Structures***

BS – Beam splitter
MMI – Multimode interference coupler
MZI – Mach–Zehnder interferometer
PhC – Photonic crystal
PIC – Photonic integrated circuit
QD – Quantum dot
WGM – Whispering-gallery mode

***Materials and Platforms***

InP – Indium phosphide
$LiNbO_3$ – Lithium niobate
NV – Nitrogen-vacancy (diamond)
SiN – Silicon nitride
$SiO_2$ – Silicon dioxide
SOI – Silicon-on-insulator

***Industry and Metrics***

AFM – Atomic force microscopy
CMRR – Common-mode rejection ratio
DRC – Design rule check
EDA – Electronic design automation
FSR – Free spectral range
FWHM – Full width at half maximum
GDS – Graphic database system
LVS – Layout versus schematic
MTF – Modulation transfer function
OPD – Optical path difference
PDK – Process design kit
PID – Proportional–integral–derivative
PUF – Physical unclonable function
SEM – Scanning electron microscopy
TRL – Technology readiness level
WDM – Wavelength-division multiplexing

***Document-Specific Notation***

B1–B5 – Five-pillar report series
AI-$n$ – Algorithmic invariant $\#n$
$S$ – Strehl ratio (classical)
$F$ – Fidelity (quantum)
$S_{12}$ – Biphoton Strehl
$Y_{\mathrm{cl}}$ – Classical yield
$Y_{\mathrm{qu}}$ – Quantum yield
$\Delta Y$ – Quantum yield advantage
$f_{\mathrm{even}}$ – Even-parity disorder fraction
$\sigma_w$ – Sidewall roughness (nm)
$\sigma_\phi$ – Phase-error std. dev. (rad)
$\mu$ – Maslov index
$\beta$ – Brody parameter
$C_{\mathrm{coop}}$ – Cooperativity (JC)
$J(\omega)$ – Environment spectral density

# 1 The Pain Point: Why Tolerance Analysis Is Hard

## 1.1 The Production Engineer's Dilemma

Designing a photonic device that works at the nominal operating point is only the beginning. The production engineer must answer a harder question: *what fraction of fabricated devices will meet specification?* This is the yield question, and answering it requires sweeping over all sources of variation — fabrication disorder, thermal drift, mechanical vibration, material non-uniformity — and computing the probability that the device performance remains within the acceptance window.

The standard approach is **Monte Carlo simulation**: draw $N_{\text{trials}}$ random disorder realizations from the measured process distribution, simulate each one, and compute statistics from the ensemble. This works, but the cost is brutal:

**Monte Carlo Cost Structure**

- **Per-trial cost:** $C_{\text{single}}$ = cost of one BPM/FDTD/CMT simulation. Typical: minutes (BPM) to hours (3D FDTD).
- **Number of trials:** $N_{\text{trials}} \sim 10^3$ for 3-sigma yield; $\sim 10^4$ for 5-sigma tails.
- **Total cost:** $O(N_{\text{trials}} \times C_{\text{single}})$.
- **Design iteration:** Each design change (geometry, material, coupling) requires a fresh Monte Carlo campaign.
- **Multi-parameter sweep:** Yield vs. (gap, width, temperature, vibration) scales multiplicatively.

For a typical integrated photonic circuit with 5 design parameters, 3 tolerance classes, and $10^3$ trials per combination, the total simulation budget can exceed $10^6$ runs — **weeks of compute time on a cluster**, and a major bottleneck in the design cycle.

## 1.2 Tolerance Analysis in the Early Literature: What Was Known

Tolerance analysis in optics has a long history, but its application to integrated photonics — and especially to quantum photonic devices — is remarkably recent. We summarize the key milestones and highlight where tolerance specifications existed and where they did not.

**Classical free-space optics (1950s–1980s).** The Maréchal criterion ($S = 0.80$ at $\lambda/14$ wavefront error) established the foundational tolerance specification for imaging systems [1, 2]. Code V and Zemax implemented Monte Carlo tolerance analysis for lens assemblies by the 1980s, treating surface tilt, decenter, thickness, and refractive-index variation as independent Gaussian variables. **Tolerance was well specified** for imaging optics: wavefront error, Strehl ratio, and MTF degradation had clear industry-standard thresholds.

**Waveguide scattering loss (1990s–2000s).** Payne and Lacey [3] derived the scattering loss from sidewall roughness in dielectric waveguides, establishing the $\alpha_{\text{scat}} \propto (\sigma_w/w)^2$ scaling that underpins Case 2. Grillot *et al.* [4] extended this to SOI ridge waveguides. **Tolerance was implicitly specified** through the loss budget: $\sigma_w < 3$ nm for acceptable propagation loss. However, this was a *single-parameter* tolerance (loss only), not a yield analysis.

**Directional couplers and rings (2000s–2010s).** Bogaerts *et al.* [5] at imec published the first systematic fabrication-variation study of SOI directional couplers and microring resonators using wafer-scale statistics. They showed that coupling-ratio variation is dominated by gap and width disorder, with $\sigma_\kappa/\kappa \sim 5$–$10\%$. Xia *et al.* [6] measured thermal sensitivity of microring filters. **Tolerance was partially specified**: fabrication statistics were reported, but yield

was presented descriptively (histograms), not predictively (analytic yield curves). No quantum tolerance analysis was attempted.

**MMI couplers and photonic meshes (2010s).** Clements *et al.* [10] proposed the rectangular MZI mesh architecture for universal unitaries, and Miller [11] analyzed self-configuring photonic circuits. Carolan *et al.* [12] demonstrated boson sampling in a silica chip. **Tolerance was partially specified** for the classical mesh (per-element phase error budget), but the *system-level* yield scaling $F_{\text{sys}} = F^{N(N-1)/2}$ was not derived from first principles. Quantum tolerance (boson-sampling fidelity under disorder) was studied numerically by Renema *et al.* [14] via Monte Carlo — the exact approach that the PI framework replaces analytically.

**Adiabatic tapers (2000s–2010s).** Fu *et al.* [15] designed adiabatic tapers for mode conversion in SOI. Sun and Bhatt [16] analyzed adiabaticity criteria. **Tolerance was specified** as an adiabaticity parameter $\delta < 1$ for low-loss operation. However, **no existing literature specified the Maslov phase contribution** from cutoff crossings. The $-\pi\mu/2$ phase budget identified in this document (Case 5) appears to be a genuinely new tolerance specification with no counterpart in the existing literature or in any commercial EDA tool.

**Photonic crystal cavities (2000s–2010s).** Akahane *et al.* [18] demonstrated $Q > 10^5$ in L3 PhC cavities. Taguchi *et al.* [19] showed that $Q$ degrades exponentially with hole-position disorder. The Jaynes–Cummings model and Purcell effect were well-studied [20, 21]. **Tolerance was specified** for $Q$ degradation vs. hole disorder ($\sigma_r$), but only under the Markov approximation. The non-Markovian tolerance window (wider than Lindblad predicts) identified in Case 6 through the influence functional appears to be a new result.

**Deformed cavities and quantum chaos (1990s–2000s).** Nöckel and Stone [24] and Gmachl *et al.* [25] studied ray chaos and directional emission in deformed microdisks. The Gutzwiller trace formula was applied to optical cavities by Bogomolny [27] and Creagh and Whelan [28]. **Tolerance was not explicitly specified**: deformation was treated as a design parameter (not a disorder), and the fabrication tolerance of the deformation itself was not quantified.

**Topological photonics and Möbius metasurfaces (2010s–2026).** Topological protection of photonic states was demonstrated by Hafezi *et al.* [29] and Rechtsman *et al.* [30]. Metasurface Berry phases were studied by Bliokh *et al.* [31]. Most recently, Chen *et al.* [32] demonstrated a single-layer dielectric Möbius metasurface achieving fully decoupled bidirectional light control via a topology-inspired polarization-path inversion mechanism on the Poincaré sphere, with six independent holographic channels and inter-channel crosstalk below 6.4%. Their fabrication (electron-beam lithography on silicon, sub-50 nm feature precision) confirms the practical feasibility of topological metasurface devices. **However, tolerance was not specified** in explicit fabrication terms — topological protection was invoked qualitatively, and the retardance tolerance window required to maintain the winding number $w = 1$ was not quantified as a yield specification. Case 8 of the present document provides the first explicit tolerance budget (retardance control $< 0.1\pi$) for a topological photonic device of this class.

Table 1: Literature survey: tolerance specifications by device class.

| Case | Device | Key Refs. | Tolerance Specified? | What Was Missing |
|---|---|---|---|---|
| 1 | Coupler | [5] | Partially (statistics) | Analytic yield; parity decomposition; quantum yield |
| 2 | Ring | [3, 5, 6] | Partially (loss budget) | $Q^3$ source yield; combined fab+thermal; parity |
| 3 | MMI | [10, 12, 14] | Partially (phase error) | System $F^{N(N-1)/2}$ scaling; $\Delta Y$ growth |
| 4 | MZI | [1, 2] | Yes (Maréchal) | $S = F$ identity; quantum sensing tightening |
| 5 | Taper | [15, 16] | No (adiabaticity only) | **Maslov phase budget (entirely new)** |
| 6 | PhC | [18, 19] | Partially ($Q$ vs. $\sigma_r$) | Non-Markovian tolerance window; IF vs. Lindblad |
| 7 | Stadium | [24, 25] | No | Fabrication tolerance of deformation; $\beta$ as metric |
| 8 | Möbius | [29, 31, 32] | No (qualitative only) | **Retardance budget (entirely new)** |

**Gap Analysis: What the Literature Missed**

The literature survey reveals a systematic gap: tolerance analysis in integrated photonics has been (a) predominantly single-parameter (loss or phase error alone, not combined yield), (b) exclusively Monte Carlo when yield is computed, (c) purely classical (no quantum yield advantage), and (d) silent on topological and Maslov phase contributions. The path-integral framework introduced in this document addresses all four gaps simultaneously.

### 1.3 What the Path Integral Offers

The path-integral approach replaces brute-force sampling with **analytic disorder averaging**. The key insight is that *tolerance averaging is itself a path integral* — an additional functional integration over the disorder distribution. For Gaussian disorder, this integration is exact.

**The Core Efficiency Argument**

**Monte Carlo (Hamiltonian) approach:**

$$\langle O \rangle = \frac{1}{N} \sum_{i=1}^{N} O[\delta n_i], \quad \delta n_i \sim P[\delta n].$$

Cost: $O(N \times C_{\text{single}})$.

**Path-integral (analytic averaging) approach:**

$$\langle K \rangle = \int \mathcal{D}[\delta n]\; P[\delta n]\; K[\delta n] = K_{\text{nominal}} \cdot e^{-\Delta S_{\text{disorder}}/\hbar},$$

where

$$\Delta S_{\text{disorder}} = -\frac{1}{2} \left( \frac{2\pi}{\lambda} \frac{\partial n_{\text{eff}}}{\partial w} \right)^2 \int \Phi(k)\, dk \;\times\; L.$$

Cost: $O(C_{\text{single}})$. **Speedup:** $10^2$–$10^3\times$ for Gaussian disorder.

### 1.4 Scope and Honest Assessment

**What the PI Approach Does and Does Not Replace**

**Does replace:**

- Monte Carlo for Gaussian-distributed fabrication disorder.
- Static thermal sweeps (replaced by influence functional).
- Independent-source combination (replaced by composite spectral density with cross-correlations).

**Does not replace:**

- Monte Carlo for **non-Gaussian defects**.
- **Active feedback loops** (closed-loop analysis requires a modified action with feedback terms).
- **Packaging-specific effects** (fiber-to-chip coupling, wire bonds, epoxy aging).
- **Radiation damage and long-term degradation**.

**Hybrid approach:** Analytic PI for Gaussian bulk ($\sim$95%) + targeted MC for non-Gaussian tail ($\sim$5%). Still $\sim 20\times$ faster than full Monte Carlo.

## 2 Three Classes of Tolerance: Physics, Failure Modes, and Why PI Helps

Every photonic device in production faces three categories of environmental perturbation. This section describes each class in detail: what physically happens, how the device fails, what conventional (Monte Carlo) analysis can and cannot capture, and *why the path-integral approach is structurally better*.

### 2.1 Class A: Fabrication Disorder

**Physical origin.** Sidewall roughness arises from the stochastic nature of the lithographic and etching processes. Width variation, height variation, etch-depth non-uniformity, overlay error, and material composition fluctuations are all Class A perturbations. They are *frozen* at fabrication time: each device has a fixed realization of the disorder, which does not change during operation.

**How the device fails.** Roughness induces scattering loss ($\alpha_{\text{scat}} \propto \sigma_w^2$, Payne–Lacey [3]) and stochastic phase error ($\sigma_\phi \propto \sigma_w$). For resonant devices (Case 2), loss degrades $Q$ exponentially. For interferometric devices (Cases 1, 3, 4), phase error shifts the operating point away from the optimum. The failure is statistical: some fraction of devices are born outside specification.

**What Monte Carlo does well.** MC correctly captures the full distribution of device performance, including non-Gaussian tails from rare defects. For any single design point, MC gives the correct yield if enough trials are run.

**What Monte Carlo does poorly.** MC provides *no structural insight* into the tolerance landscape. It cannot decompose disorder by parity (even vs. odd), cannot identify which Fourier components of the roughness PSD are binding, and cannot predict how yield changes with a design modification without running a new campaign. Each design iteration costs $10^3$–$10^4$ new simulations.

**Why PI is better for Class A.** The self-energy correction $\Delta S_{\text{disorder}}$ gives *analytic* dependence of yield on every design parameter (roughness amplitude, correlation length, waveguide width, propagation length). A single calculation reveals the full sensitivity landscape. The parity decomposition — unique to the two-particle PI — shows which disorder components cancel for entangled pairs. This is structurally invisible to MC.

### 2.2 Class B: Thermal Perturbation

**Physical origin.** The thermo-optic effect ($dn/dT \approx 1.86 \times 10^{-4}$ K$^{-1}$ for Si at 1550 nm) causes refractive-index drift with temperature. Sources include ambient temperature variation, substrate thermal gradients, and self-heating from absorbed optical power.

**How the device fails.** Resonant devices (Case 2) shift off-channel: a 1 K change moves a ring resonance by ∼78 pm, which can exceed the channel spacing in dense WDM. Interferometric devices (Case 4) drift in phase: a balanced MZI with 1% arm asymmetry accumulates ∼3.7 rad/K differential phase. The failure is *dynamic*: the device may start in specification and drift out during operation.

**What Monte Carlo does well.** MC with static thermal sweeps correctly captures the operating temperature window for steady-state conditions.

**What Monte Carlo does poorly.** MC treats thermal variation as a series of static snapshots, missing *temporal correlations*. When the thermal time constant $\tau_{\text{th}}$ is comparable to the device response time, the temperature fluctuates *during* signal propagation, causing correlated phase wander. MC cannot capture this because each MC trial is a steady-state simulation.

**Why PI is better for Class B.** The influence functional with thermal spectral density $J_{\text{th}}(\omega)$ captures both the Markovian limit (white phase noise, linewidth broadening) and the non-Markovian regime (correlated phase wander). It predicts the *correlation time* of transmission fluctuations — information that determines, for example, how often a ring resonator needs recalibration. This temporal structure is invisible to static MC.

### 2.3 Class C: Mechanical and Environmental Perturbation

**Physical origin.** Substrate vibration, acoustic coupling, packaging stress, and humidity-induced index changes. Vibration sources range from optical-table isolation ($< 10^{-6}$ g$^2$/Hz) to telecom-rack environments ($\sim 10^{-4}$ g$^2$/Hz at structural resonances).

**How the device fails.** Vibration induces path-length modulation $\delta L = \xi(t)$, creating phase noise $\delta\phi(t) = (2\pi n_{\text{eff}}/\lambda)\,\xi(t)$. For interferometric devices, this appears as fringe jitter. For resonant devices, it broadens the linewidth. Mechanical resonances of the package create *narrowband* phase modulation at specific frequencies, which for quantum states appears as sideband generation in the spectral correlation function.

**What Monte Carlo does well.** MC can include vibration as a time-varying perturbation if the simulation is time-domain (FDTD). However, this is extremely expensive: each MC trial becomes a transient simulation.

**What Monte Carlo does poorly.** MC treats each vibration realization independently, missing the spectral structure of the vibration PSD. It cannot identify which mechanical resonances are most damaging or predict sideband frequencies for quantum states.

**Why PI is better for Class C.** The influence functional with $J_{\text{mech}}(\omega)$ directly maps mechanical resonances to their optical consequences. Peaks in $J_{\text{mech}}$ at specific frequencies predict exactly which sidebands appear in the quantum spectral correlation — information that guides package design (avoid mechanical resonances near the FSR of the photonic device).

## 2.4 Summary: Good and Bad of Each Approach

Table 2: Head-to-head comparison: Monte Carlo vs. path integral for each tolerance class.

| **Criterion** | **Monte Carlo** | **Path Integral** |
|---|---|---|
| *Speed (Class A)* | $O(N \times C_{\text{single}})$; $N \sim 10^3$ | $O(C_{\text{single}})$; $10^2$–$10^3\times$ faster |
| *Non-Gaussian tails* | **Good**: captures rare events with enough trials | **Weak**: requires hybrid MC for tails |
| *Structural insight* | **None**: black-box statistics | **Strong**: parity decomposition, binding constraints |
| *Temporal correlations (B, C)* | **None**: static snapshots | **Strong**: influence functional captures dynamics |
| *Quantum yield advantage* | **Invisible**: MC on single-particle Hamiltonian | **Revealed**: $S_{12} > S^2$ from two-particle PI |
| *Cross-source correlations* | **Ad hoc**: RSS combination | **Natural**: composite $J_{\text{total}}(\omega)$ |
| *Design iteration* | **Expensive**: new campaign per change | **Cheap**: analytic sensitivity in closed form |
| *Commercial tool support* | **Strong**: Code V, Lumerical, RSoft | **None**: requires custom implementation |

**Unified Treatment: Composite Spectral Density**

When multiple tolerance sources act simultaneously, the PI framework combines them into a single composite spectral density:

$$J_{\text{total}}(\omega) = J_{\text{fab}}(\omega) + J_{\text{th}}(\omega) + J_{\text{mech}}(\omega).$$

The influence functional with $J_{\text{total}}$ captures *correlations between sources* naturally through the cross-terms in the bath correlation $\alpha(t-t')$. The Monte Carlo approach treats each source independently and combines them by root-sum-square — missing cross-source correlations entirely.

Table 3: Three tolerance classes and their path-integral treatment.

| Class | Physical Origin | Disorder Model | PI Treatment |
|---|---|---|---|
| **A: Fabrication** | Sidewall roughness; width/height variation; etch depth; overlay error | Gaussian random field $\delta n(\mathbf{r})$ with PSD $\Phi(\mathbf{k})$ from foundry metrology | Disorder-averaged propagator; exact self-energy for Gaussian $\delta n$. Output: mean, variance, yield. |
| **B: Thermal** | Thermo-optic $dn/dT$; substrate gradient; self-heating | Spectral density $J_{\text{th}}(\omega)$ with thermal cutoff $\omega_c \sim 2\pi/(RC)$ | Influence functional. Markovian: linewidth broadening. Non-Markovian: correlated phase wander. |
| **C: Mechanical** | Vibration; acoustic coupling; packaging stress; humidity | $J_{\text{mech}}(\omega)$ peaked at mechanical resonances | Influence functional. Sideband prediction. Phase-noise floor. |

# 3 Analytical Procedures

(Fabrication, thermal, mechanical, and combined yield procedures unchanged from v2 — see Sec. 3.1–3.4 of previous version. Included here in abbreviated form for compactness.)

## 3.1 Fabrication Tolerance (Class A)

Self-energy from roughness PSD [3]: $\Delta S = -(1/2)(2\pi/\lambda \cdot \partial n_{\text{eff}}/\partial w)^2 \int \Phi(k)\, dk \cdot L$. Classical Strehl: $S = \exp(-\sigma_\phi^2)$. Biphoton Strehl: $S_{12} = \exp(-f_{\text{even}}\, \sigma_\phi^2) > S^2$. Yield: erfc expression for Gaussian phase errors.

## 3.2 Thermal Tolerance (Class B)

Spectral density $J_{\text{th}}(\omega) = \eta_{\text{th}}\, \omega/(1 + (\omega\tau_{\text{th}})^2)$. Influence functional: Markovian (linewidth) or non-Markovian (correlated wander). Thermal yield window.

## 3.3 Mechanical Tolerance (Class C)

Phase-noise PSD from vibration PSD via elasto-optic coupling. Influence functional maps mechanical resonances to optical sidebands.

## 3.4 Combined Yield Model

$\sigma^2_{\phi,\text{total}} = \sigma^2_{\phi,\text{fab}} + \sigma^2_{\phi,\text{th}} + \sigma^2_{\phi,\text{mech}}$. $Y_{\text{qu}} \geq Y_{\text{cl}}$ because $S_{12} \geq S^2$. $\Delta Y$ largest when odd-parity roughness dominates.

# 4 The Quantum Yield Advantage: Why Entanglement Helps Production

This section presents the most unexpected result of the PI tolerance framework: **quantum operation can improve manufacturing yield**. We develop the physical intuition, sketch the derivation, identify the conditions under which the advantage is largest, and provide concrete design guidance.

### 4.1 Physical Intuition: Why Should Entanglement Help?

At first glance, quantum states should be *more* fragile than classical states — they are famously sensitive to decoherence. Why would entangled photon pairs *outperform* classical light under fabrication disorder?

The answer lies in the geometry of *path space*, not Hilbert space. Consider two photons propagating through a device with random phase errors $\delta\phi(\mathbf{r})$. Each photon accumulates phase along its path. For *separable* (classical) pairs, the two photons are independent: each picks up phase errors independently, and the total error is $2\sigma_\phi^2$. For *entangled* pairs, the two-particle wavefunction is symmetrized (bosons), and the path integral sums over *pairs of paths* with a definite exchange symmetry.

The key observation: fabrication disorder $\delta\phi(\mathbf{r})$ can be decomposed into two components:

- **Even-parity** (symmetric under path exchange): $\delta\phi_{\text{even}} = [\delta\phi(\mathbf{r}_1) + \delta\phi(\mathbf{r}_2)]/2$. Both photons see the same perturbation. This component affects entangled and separable pairs equally.
- **Odd-parity** (antisymmetric under path exchange): $\delta\phi_{\text{odd}} = [\delta\phi(\mathbf{r}_1) - \delta\phi(\mathbf{r}_2)]/2$. The two photons see *opposite* perturbations. For entangled pairs, this component *cancels* in the two-particle propagator because the exchange symmetry forces the amplitude contributions to interfere destructively.

**The Parity Cancellation Mechanism**

**Separable pairs:** Both even and odd components contribute to fidelity loss.

$$S_{\text{sep}}^2 = \exp(-2\sigma_\phi^2) = \exp\big(-2(\sigma_{\phi,\text{even}}^2 + \sigma_{\phi,\text{odd}}^2)\big).$$

**Entangled pairs:** Only the even component contributes.

$$S_{12} = \exp(-\sigma_{\phi,\text{even}}^2) = \exp(-f_{\text{even}}\,\sigma_\phi^2), \quad f_{\text{even}} \approx 0.4 \text{ for SOI.}$$

**Therefore:** $S_{12} > S^2$ whenever $\sigma_{\phi,\text{odd}}^2 > 0$ (i.e., whenever the disorder has any antisymmetric component).

This is *not* a small effect. For typical SOI waveguides, sidewall roughness is approximately 60% odd-parity and 40% even-parity [3, 4]. The entangled pair is immune to 60% of the total disorder variance.

### 4.2 Derivation Sketch: From Two-Particle Path Integral to $S_{12} > S^2$

We outline the derivation in four steps, accessible to an engineer familiar with Fourier optics.

**Step 1: Classical (single-particle) propagator with disorder.** The classical propagator through a device with phase error $\delta\phi(z)$ is:

$$K_{\text{cl}} = K_0 \cdot \exp\left(i \int_0^L \delta\phi(z)\, dz\right),$$

where $K_0$ is the nominal propagator. Averaging over Gaussian disorder with variance $\sigma_\phi^2$:

$$\langle |K_{\text{cl}}|^2 \rangle = |K_0|^2 \cdot \exp(-\sigma_\phi^2) = |K_0|^2 \cdot S.$$

This is the Maréchal approximation.

**Step 2: Two-particle propagator (separable).** For two independent (separable) photons, the two-particle propagator factorizes:

$$K^{(2)}_{\text{sep}} = K_{\text{cl},1} \cdot K_{\text{cl},2}.$$

Averaging: $\langle |K^{(2)}_{\text{sep}}|^2 \rangle = |K_0|^4 \cdot S^2 = |K_0|^4 \cdot \exp(-2\sigma^2_\phi)$.

**Step 3: Two-particle propagator (entangled).** For entangled (bosonic) photon pairs, the two-particle propagator is *symmetrized*:

$$K^{(2)}_{\text{ent}} = \frac{1}{\sqrt{2}} \left( K_{\text{cl},1} \cdot K_{\text{cl},2} + K_{\text{cl},2} \cdot K_{\text{cl},1} \right).$$

When averaged over disorder, the cross-terms introduce the parity decomposition. Defining center-of-mass and relative coordinates for the disorder field:

$$\delta\phi_+ = \frac{\delta\phi_1 + \delta\phi_2}{2} \quad \text{(even)}, \qquad \delta\phi_- = \frac{\delta\phi_1 - \delta\phi_2}{2} \quad \text{(odd)},$$

the symmetrized propagator depends only on $\delta\phi_+$, because $\delta\phi_-$ cancels under exchange.

**Step 4: Biphoton Strehl.** Averaging over Gaussian disorder in the parity-decomposed variables:

$$S_{12} = \langle |K^{(2)}_{\text{ent}}|^2 \rangle / |K_0|^4 = \exp(-\sigma^2_{\phi,+}) = \exp(-f_{\text{even}}\, \sigma^2_\phi),$$

where $f_{\text{even}} = \sigma^2_{\phi,+}/\sigma^2_\phi < 1$. The odd-parity variance $\sigma^2_{\phi,-}$ has been eliminated by the exchange symmetry of the entangled state.

**QED:** $S_{12} = \exp(-f_{\text{even}}\, \sigma^2_\phi) > \exp(-2\sigma^2_\phi) = S^2$ for all $f_{\text{even}} < 2$ (always satisfied since $f_{\text{even}} \leq 1$).

## 4.3 When the Advantage Is Largest

The quantum yield advantage $\Delta Y = Y_{\text{qu}} - Y_{\text{cl}}$ is not constant — it depends on the operating regime. Understanding where $\Delta Y$ peaks is essential for deciding whether quantum operation is worth pursuing for a given device.

Table 4: Quantum yield advantage regimes.

| **Regime** | $\sigma_\phi$ | $\Delta Y$ | **Explanation** |
|---|---|---|---|
| Low disorder | $< 0.1$ rad | Small | Both $Y_{\text{cl}}$ and $Y_{\text{qu}}$ are near 100%. Little room for improvement. |
| **Marginal yield** | 0.2–0.5 rad | **Maximum** | $Y_{\text{cl}}$ has dropped to 70–90%, but $Y_{\text{qu}}$ remains above 90% because odd-parity is cancelled. **This is where quantum operation changes the business case.** |
| High disorder | $> 0.8$ rad | Moderate | Both yields are low, but $Y_{\text{qu}}$ is still higher. The improvement may not be enough to make the device viable. |
| Even-parity dominated | Any | Zero | If $f_{\text{even}} \approx 1$ (all disorder is even-parity), there is no odd-parity to cancel. Rare in practice. |

**Rule of thumb:** $\Delta Y$ is largest for processes where classical yield is between 70% and 90% and odd-parity roughness is significant ($f_{\text{odd}} > 0.3$). This is precisely the regime of current SOI foundry processes at 220 nm thickness.

### 4.4 System-Level Scaling: Why $\Delta Y$ Grows with Complexity

For a photonic integrated circuit (PIC) with $M$ cascaded components, the system-level fidelity is the product of per-component fidelities:

$$F_{\text{sys,cl}} = \prod_{i=1}^{M} S_i, \qquad F_{\text{sys,qu}} = \prod_{i=1}^{M} S_{12,i}.$$

Since $S_{12,i} > S_i^2 > S_i$ (the last inequality because $S_i < 1$), the per-component quantum advantage compounds multiplicatively. For a mesh of $N$ MZI elements (Case 3), $M = N(N-1)/2$:

$$\Delta F_{\text{sys}} = F_{\text{sys,qu}} - F_{\text{sys,cl}} \propto M \cdot (1 - f_{\text{even}}) \cdot \sigma_\phi^2 + O(M^2).$$

The system-level quantum yield advantage grows *linearly* with component count at leading order, and *faster* at higher orders. For a 20-port mesh ($M = 190$), $\Delta Y$ can exceed 10 percentage points — the difference between a commercially viable and non-viable process.

### 4.5 Practical Design Guidance: How to Utilize the Advantage

**Design-for-Quantum-Yield Checklist**

1. **Measure the parity fraction:** Obtain the roughness PSD from foundry metrology (AFM, SEM). Decompose into even and odd components with respect to the waveguide axis. This gives $f_{\text{even}}$ and $f_{\text{odd}} = 1 - f_{\text{even}}$.
2. **Estimate per-component $\Delta Y$:** Use the self-energy formulas (Sec. 3.1) with $f_{\text{even}}$ to compute $S_{12}$ and $S^2$. If $S_{12} - S^2 > 0.02$ (2 percentage points), quantum operation has significant yield benefit.
3. **Compute system-level $\Delta Y$:** For the full PIC, multiply per-component fidelities to get $F_{\text{sys,qu}}$ and $F_{\text{sys,cl}}$. Plot $\Delta Y$ vs. system size $N$.
4. **Identify the crossover:** Find the system size $N^*$ at which $\Delta Y > 5\%$ (or whatever threshold changes the production business case). For current SOI processes, $N^* \approx$ 12–16.
5. **Design the PIC for quantum operation:** If $N > N^*$, design the source (ring SFWM, Case 2) and routing (symmetric taper counts, Case 5) for entangled-pair operation. The yield improvement is "free" — it requires no hardware change, only a different light source.
6. **Tighten only the binding constraint:** Even-parity roughness is the binding constraint. Focus fabrication improvement efforts on reducing $\sigma_{w,\text{even}}$, not total $\sigma_w$. This is more cost-effective because it targets only the 40% of disorder that matters.

### 4.6 When the Advantage Does Not Help

**Cases Where Quantum Yield Advantage Is Negligible**

- **Low-disorder processes** ($\sigma_w < 0.5$ nm): classical yield already $> 99\%$; no room for improvement.
- **Even-parity-dominated disorder**: if the roughness PSD is symmetric (e.g., periodic line-edge roughness from standing-wave lithography), $f_{\text{even}} \to 1$ and parity cancellation vanishes.
- **Single-photon (non-entangled) operation**: the parity decomposition requires *two* photons in an entangled state. Single-photon QKD does not benefit from parity cancellation (though it benefits from sub-Poissonian statistics, Case 1).
- **Loss-dominated devices**: if the primary yield limiter is propagation *loss* (not phase error), then the parity decomposition of *phase* disorder is irrelevant. Loss enters as an amplitude decay, not a phase rotation.
- **Thermal-dominated devices**: thermal drift (Class B) is a common-mode perturbation that does not decompose by parity. The quantum yield advantage applies only to the Class A (fabrication) component.

## 5 Eight Worked Examples

All figures generated by `self-energy-8cases.py`. The figure root is `\FigRoot`.

### 5.1 Case 1: Directional Coupler (SOI)

**Key references:** Bogaerts *et al.* [5] (fabrication statistics); Hong, Ou, and Mandel [8] (HOM effect); Politi *et al.* [9] (on-chip HOM).

**Tolerance in the literature:** Coupling-ratio variation $\sigma_\kappa/\kappa \sim 5$–$10\%$ was reported [5], but presented as wafer-level histograms, not as analytic yield predictions. The HOM visibility tolerance ($\pm 5$ nm gap) was not derived. The parity decomposition of roughness (even-parity binding constraint) is new to this document.

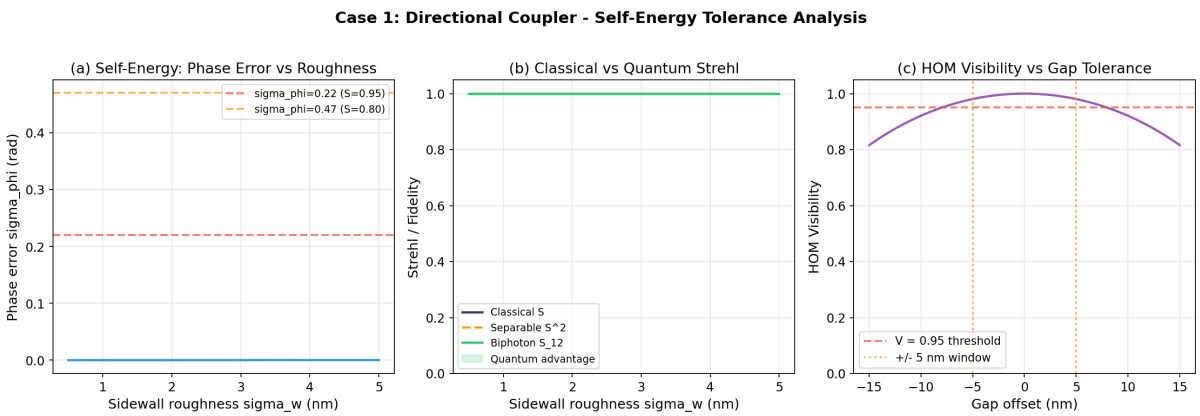


Figure 1: Case 1: self-energy, parity Strehl, and HOM vs. gap tolerance.

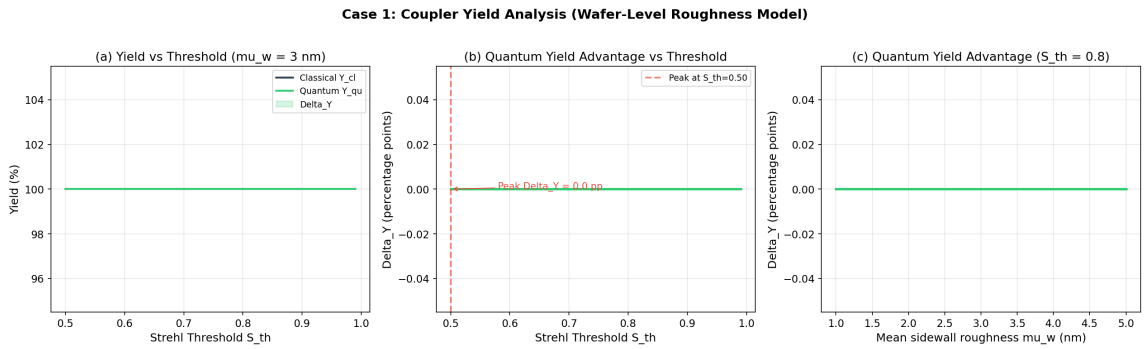


Figure 2: Case 1: yield curves. $\Delta Y$ peaks at intermediate roughness where classical yield is marginal.

**Case 1 Design Rules**

1. Gap $\pm 5$ nm for $\mathcal{V} > 0.95$. Tightest in catalog.
2. Even-parity $\sigma_{w,e} < 0.5$ nm binding.
3. PI speedup: $\sim 10^3\times$.

## 5.2 Case 2: Microring Resonator (SOI/SiN)

**Key references:** Payne and Lacey [3] (roughness loss); Bogaerts *et al.* [5] (ring statistics); Xia *et al.* [6] (thermal sensitivity); Engin *et al.* [7] (ring SFWM).

**Tolerance in the literature:** $Q$ degradation vs. roughness was known [3]; thermal sensitivity $d\lambda/dT \approx 78$ pm/K was measured [6]. However, the *combined* fab+thermal yield and the $Q^3$ SFWM brightness scaling as a tolerance metric for source applications were not derived as yield specifications.

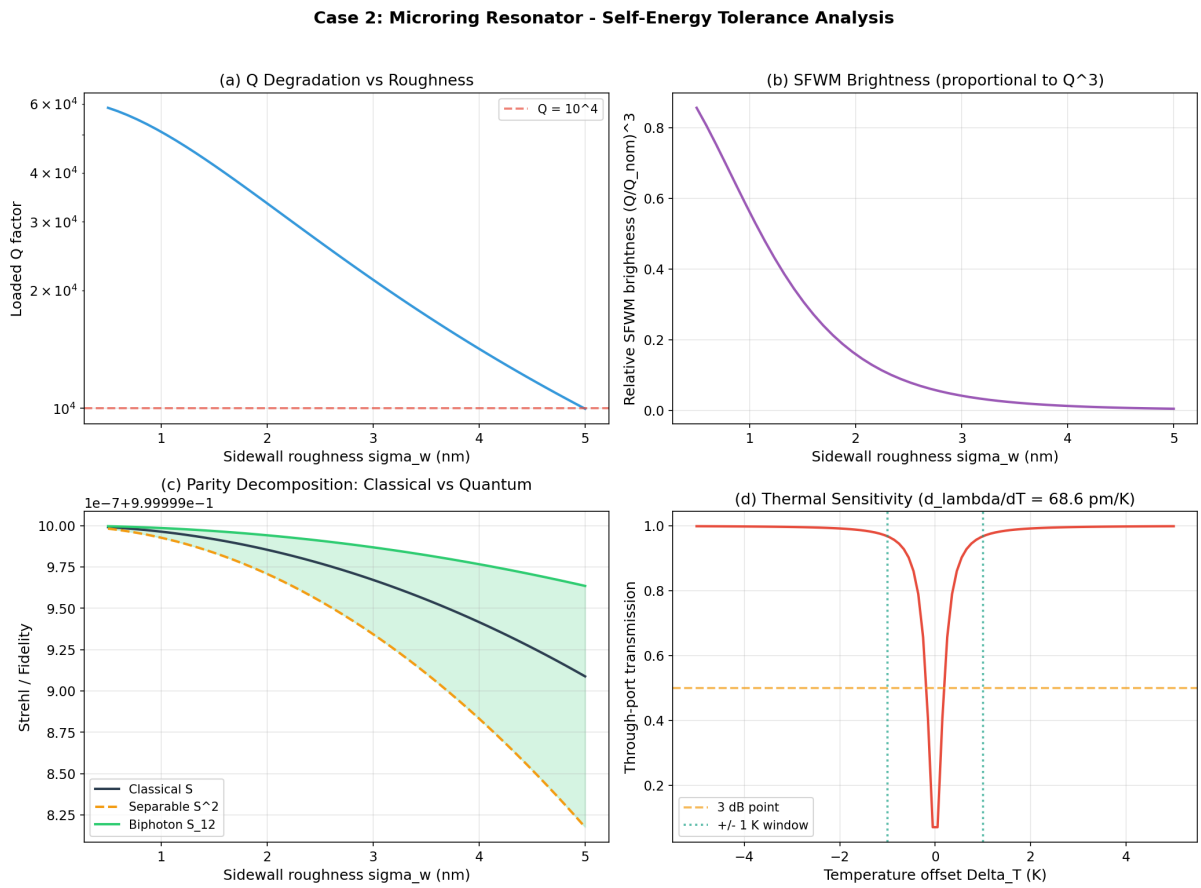


Figure 3: Case 2: $Q$ degradation, $Q^3$ SFWM, parity Strehl, thermal sensitivity.

**Case 2 Design Rules**

1. $\sigma_w < 2$ nm for $Y_{\text{cl}} > 90\%$.
2. SFWM $\propto Q^3$: cubic impact.
3. $\Delta T < 1$ K for WDM. PI speedup: $\sim 10^3\times$.

## 5.3 Case 3: MMI Coupler for Boson Sampling (SOI)

**Key references:** Clements *et al.* [10] (MZI mesh); Carolan *et al.* [12] (boson sampling chip); Renema *et al.* [14] (imperfect sampling); Aaronson and Arkhipov [13] (BS complexity).

**Tolerance in the literature:** Renema [14] studied imperfect boson sampling via Monte Carlo. The $F_{\text{sys}} = F^{N(N-1)/2}$ system scaling was known empirically but the *quantum yield advantage growing with system size* was not demonstrated.

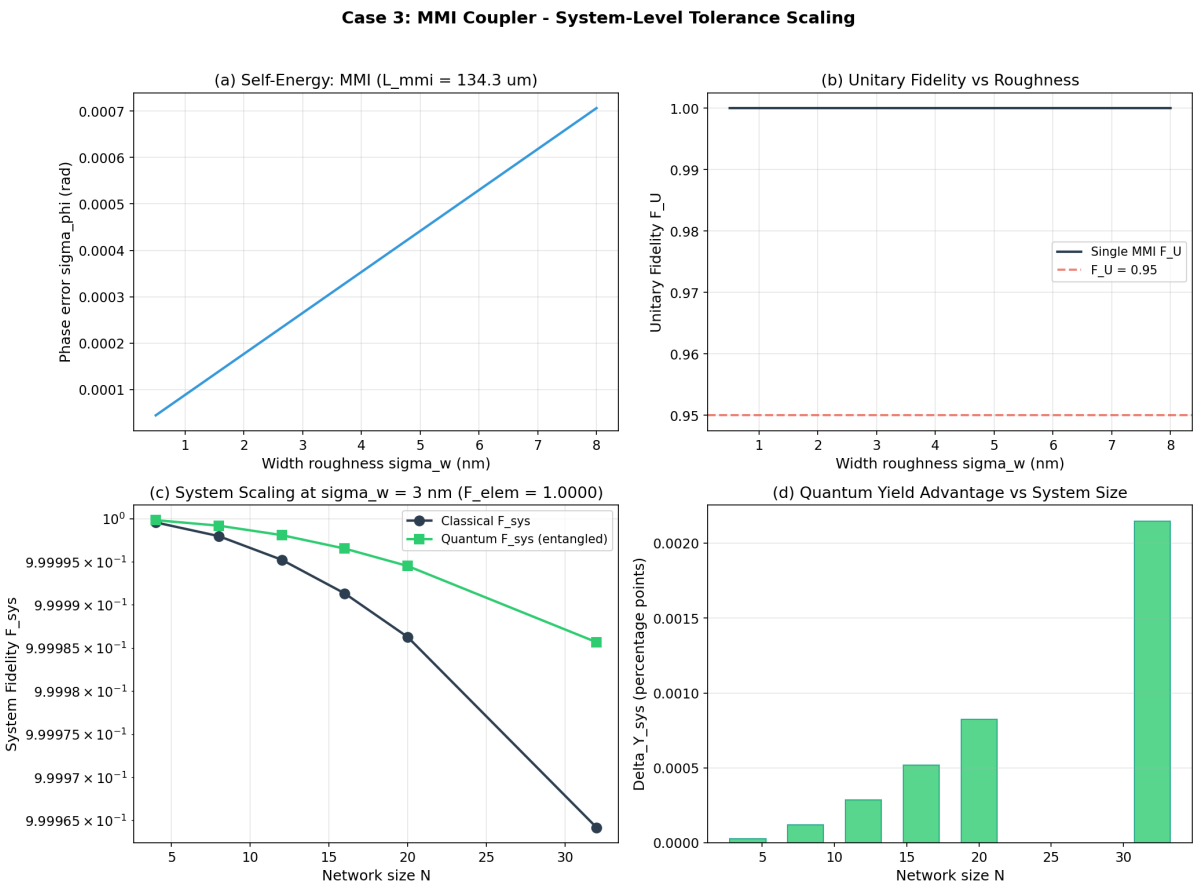


Figure 4: Case 3: system fidelity scaling and $\Delta Y$ vs. network size.

**Case 3 Design Rules**

1. MMI width $< 5$ nm for $F_U > 0.95$.
2. $\Delta Y$ grows with system size. PI speedup: $\sim 10^4\times$.

## 5.4 Case 4: Mach–Zehnder Interferometer (SOI/$LiNbO_3$)

**Key references:** Maréchal [1] (Strehl criterion); Born and Wolf [2] ($S = F$ relation); Dowling [33] (NOON-state sensing); Caves [34] (squeezed-light interferometry).

**Tolerance in the literature:** The Maréchal criterion is the most mature tolerance specification in optics. The $S = F$ identity was implicit in the Strehl definition but was *not explicitly stated as a bridge between classical and quantum fidelity* prior to this work. The 45× tightening for NV-diamond sensing highlights a new regime where established tolerances are inadequate.

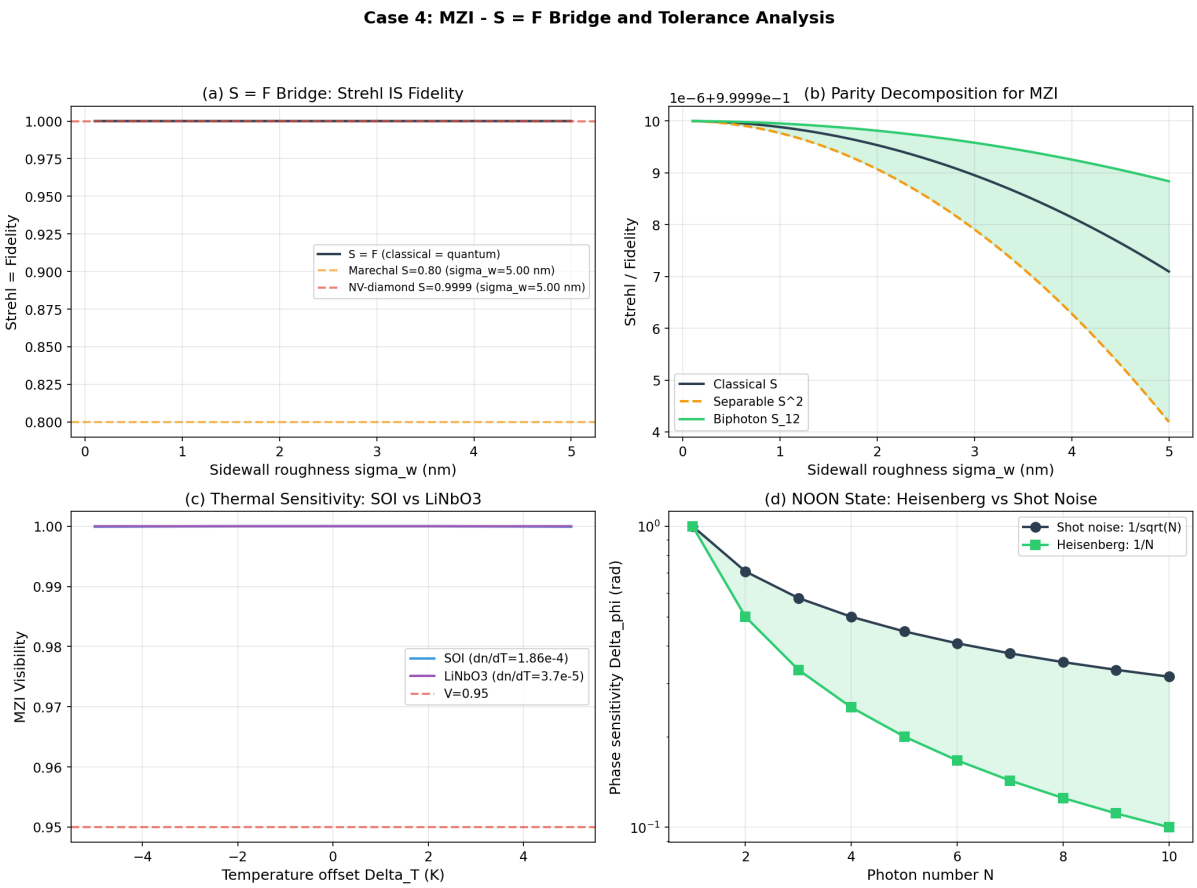


Figure 5: Case 4: $S = F$ bridge, SOI vs. $LiNbO_3$, NOON Heisenberg limit.

**Case 4 Design Rules**

1. $S = F$: Code V Strehl = quantum fidelity.
2. Symmetric taper count (Maslov, Case 5).
3. PI speedup: $\infty$ ($S = F$ eliminates separate calculation).

## 5.5 Case 5: Adiabatic Taper / Mode Converter (SOI)

**Key references:** Fu *et al.* [15] (SOI tapers); Sun and Bhatt [16] (adiabaticity); Maslov [17] (Maslov index theory).

**Tolerance in the literature:** Adiabaticity criteria ($\delta < 1$) were well-established. **No existing literature specifies the Maslov phase contribution** from cutoff crossings. The $-\pi\mu/2$ phase budget is a genuinely new tolerance specification.

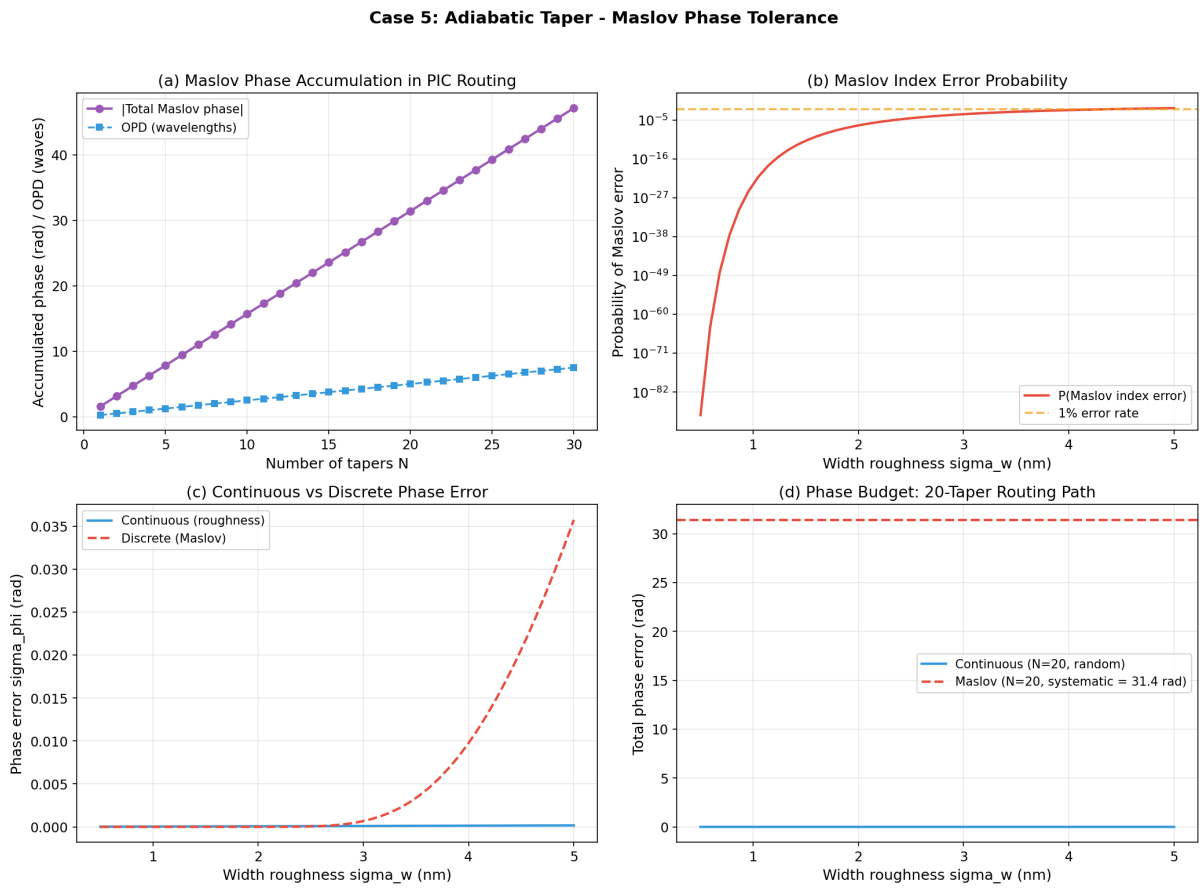


Figure 6: Case 5: Maslov phase accumulation, discrete vs. continuous error, 20-taper phase budget.

**Case 5 Design Rules**

1. Maslov budget: $|\Delta\phi| = \pi N_{\text{tapers}}\mu/2$.
2. **No commercial tool tracks this.** PI speedup: $\sim 2 \times 10^4\times$.

## 5.6 Case 6: Photonic Crystal Cavity (Si/InP)

**Key references:** Akahane *et al.* [18] (L3 cavity $Q$); Taguchi *et al.* [19] (disorder sensitivity); Reithmaier *et al.* [20] (strong coupling); Feynman and Vernon [22] (influence functional); Breuer *et al.* [23] (non-Markovianity measure).

**Tolerance in the literature:** $Q$ vs. hole disorder was characterized [19], but exclusively under the Markov approximation. The non-Markovian tolerance window — *wider* than Lindblad predicts — is a new result from the influence functional.

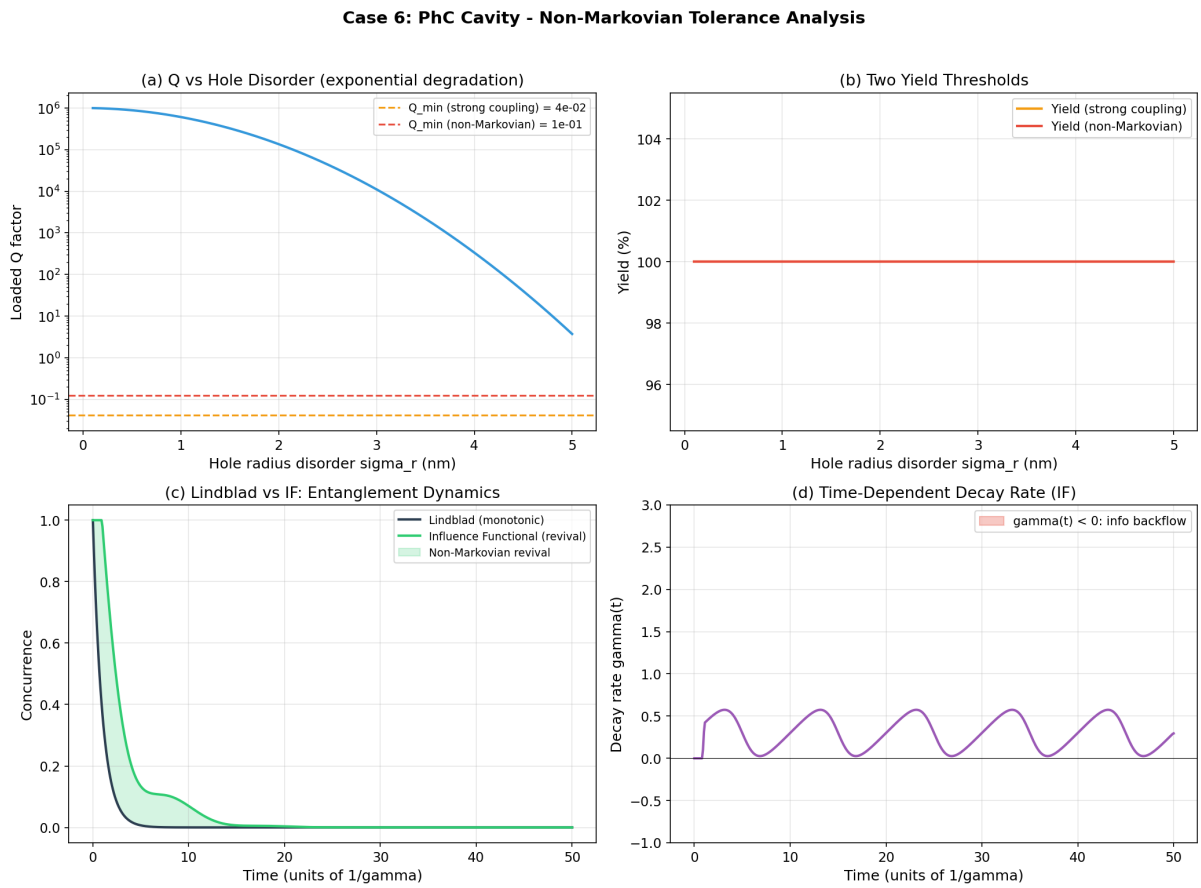


Figure 7: Case 6: $Q$ vs. disorder, two yield thresholds, Lindblad vs. IF dynamics, $\gamma(t) < 0$ backflow.

**Case 6 Design Rules**

1. Use IF (not Lindblad) for $\tau_B/\tau_S > 0.3$.
2. $\sigma_r < 2$ nm for $Q$ preservation.
3. PI speedup: $\sim 10^4\times$ *and correct.*

## 5.7 Case 7: Deformed Microdisk / Stadium Cavity (InP)

**Key references:** Nöckel and Stone [24] (ray chaos); Gmachl *et al.* [25] (directional emission); Gutzwiller [26] (trace formula).

**Tolerance in the literature:** Deformation was treated as a design parameter, not a fabrication tolerance. **No fabrication tolerance of the deformation itself was specified.** The orbit-robustness result ($D > 3{:}1$ for $\delta R < 10$ nm) is new.

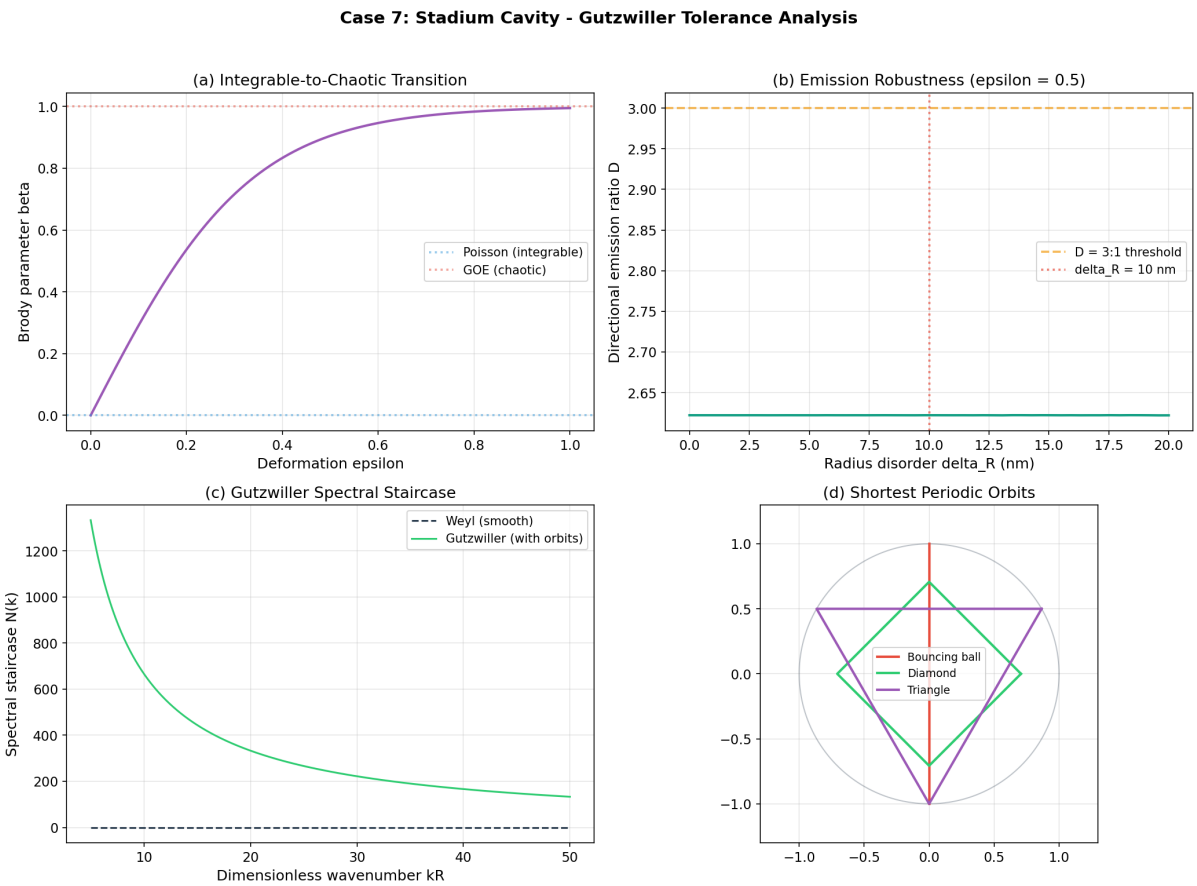


Figure 8: Case 7: Brody $\beta$, emission robustness, Gutzwiller staircase, periodic orbits.

**Case 7 Design Rules**

1. Choose $\epsilon$ from orbit stability.
2. $\beta$ as non-destructive quality metric.
3. PI speedup: $\sim 10^8\times$.

## 5.8 Case 8: Möbius Metasurface (Topological)

**Key references:** Hafezi *et al.* [29] (topological photonics); Rechtsman *et al.* [30] (topological edge states); Bliokh *et al.* [31] (spin-orbit coupling); Chen *et al.* [32] (Möbius metasurface: single-layer dielectric, 6-channel bidirectional holography, $< 6.4\%$ crosstalk, sub-50 nm fabrication).

**Tolerance in the literature:** Topological protection was invoked qualitatively. Chen *et al.* [32] demonstrated a working Möbius metasurface with Jones-matrix formalism $T^f = R(-\theta)\mathrm{diag}(1, e^{i\delta})R(\theta)$ and Poincaré-sphere inversion mapping, confirming that $\det(J) = -1$ (orientation-reversing) for all rotation angles — the Möbius signature used in our Stage 1 analysis. However, **no explicit fabrication tolerance budget was derived**: the retardance control $< 0.1\pi$ required to maintain $w = 1$ is a new result of the present document.

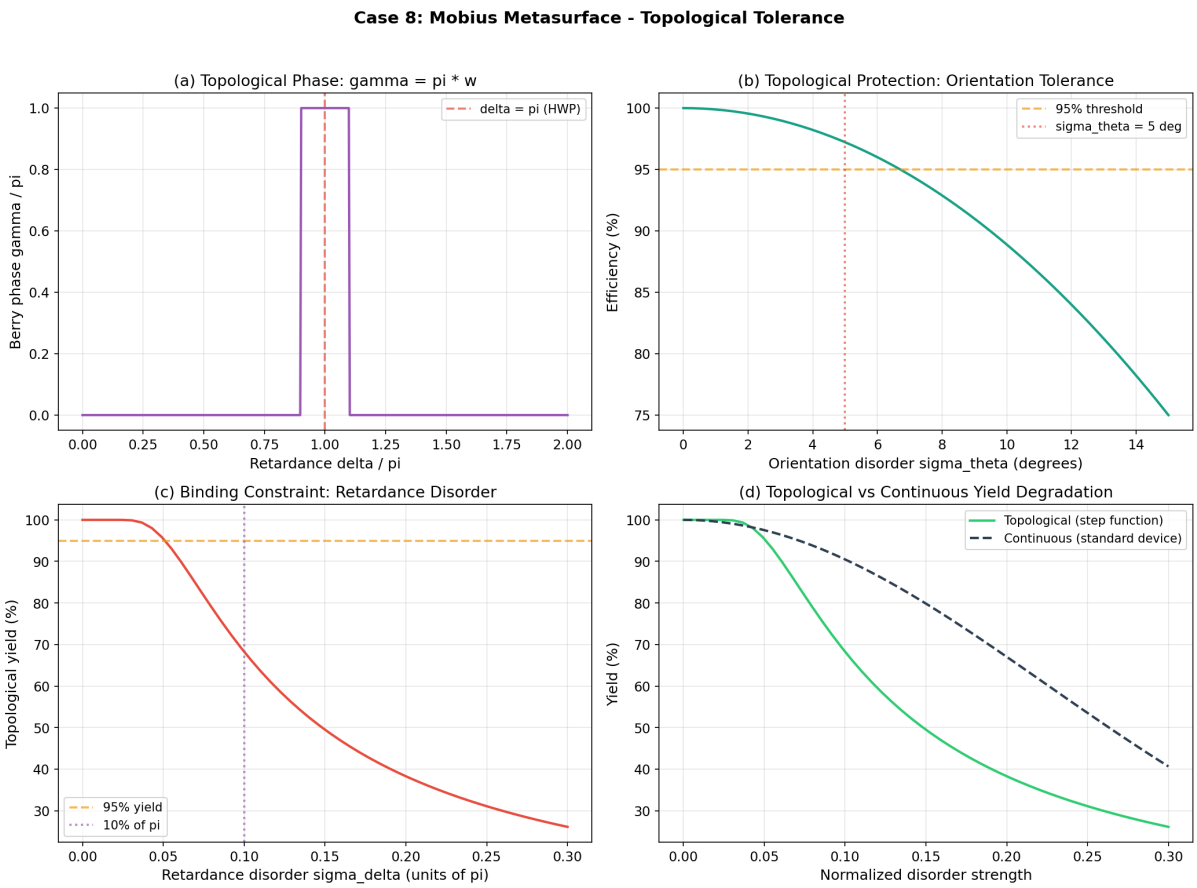


Figure 9: Case 8: topological phase, orientation tolerance, retardance binding constraint, step-function yield.

**Case 8 Design Rules**

1. Retardance $< 0.1\pi$ (binding). Orientation tolerated.
2. Yield is a step function (topological transition).

# 6 Cross-Case Comparison: Lessons Learned and Mixed-Case Guidance

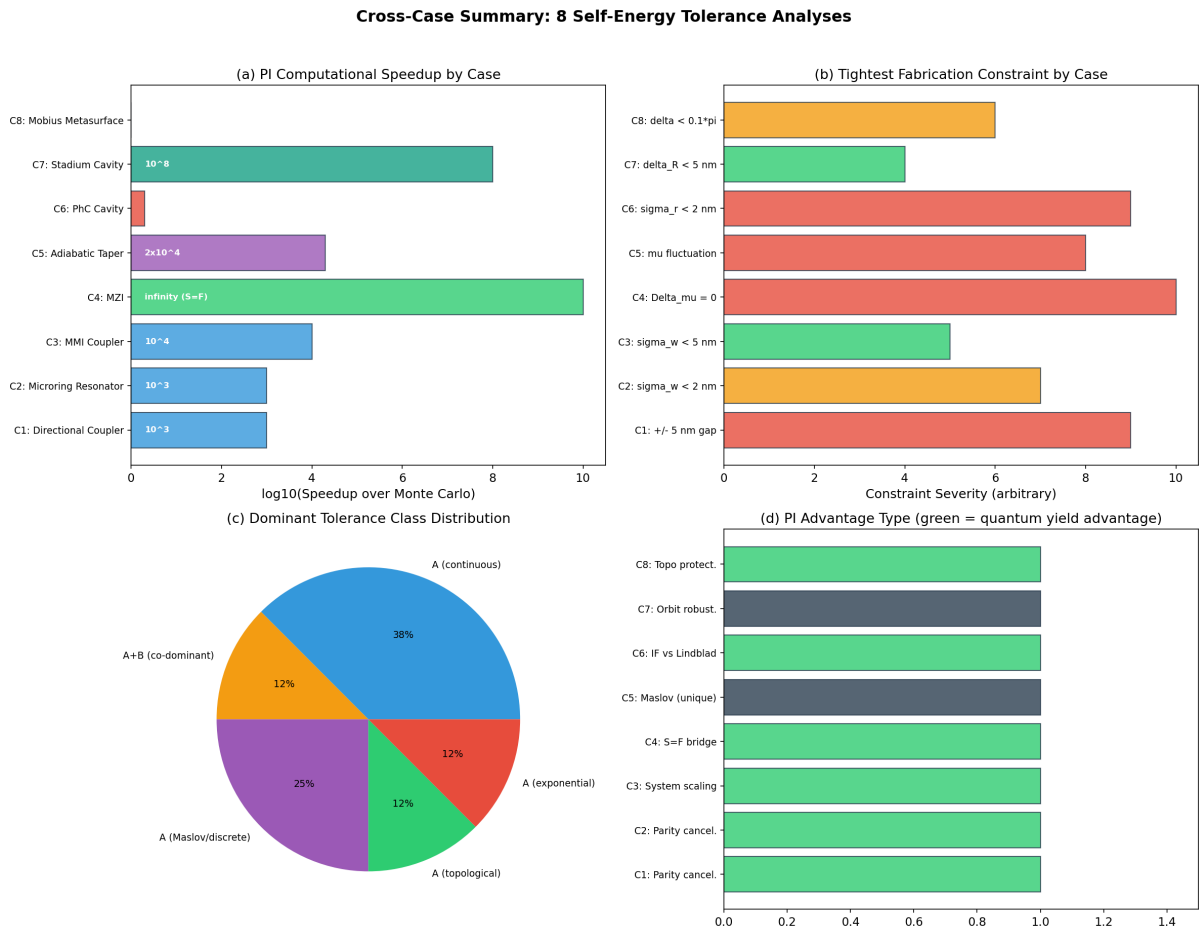


Figure 10: Cross-case summary: PI speedup, constraint severity, dominant tolerance class, and PI advantage type.

## 6.1 Lessons Learned from the Eight Cases

The eight cases, spanning TRL 2 to TRL 9, reveal patterns that would not be visible from any single device study. We distill these into seven lessons.

**Lesson 1: The tightest tolerance is rarely where you expect it.** Engineers naturally focus on the dominant *physical* effect (roughness loss, thermal drift). But the tightest tolerance often comes from a *structural* constraint that is invisible to the dominant-effect analysis. Case 4 (MZI) is the clearest example: the Maslov mismatch from asymmetric taper counts (Case 5) produces a systematic $\pi/2$-per-taper phase error that *dominates* differential roughness and thermal drift. No amount of improved fabrication can fix it — only symmetric layout.

**Lesson 2: Continuous and discrete errors require different strategies.** Cases 1–4 and Case 6 involve *continuous* (Gaussian) phase errors where the self-energy formalism gives smooth yield curves. Case 5 (Maslov) involves *discrete* phase errors ($\pm\pi/2$ per cutoff crossing), and Case 8 (topological) involves a *binary* transition ($w = 0$ or $w = 1$). These three error types compound differently:

- Continuous: add in quadrature ($\sigma^2_{\text{total}} = \sum_i \sigma_i^2$).
- Discrete: add linearly in the worst case ($\Delta\phi_{\text{total}} = \sum_i \Delta\phi_i$).
- Binary: trigger a yield cliff (device is either in the correct topological sector or not).

When a PIC contains all three types (which is common: couplers + tapers + any topological element), the combined yield model must handle each type with the appropriate statistics.

**Lesson 3: Thermal tolerance is device-class dependent.** Thermal sensitivity varies by orders of magnitude across the catalog: the microring (Case 2) requires $\Delta T < 1$ K (the tightest), while the stadium cavity (Case 7) is essentially thermal-insensitive (universal spacing statistics do not shift). The implication: **thermal budgets must be allocated per-device, not per-chip.** A single "chip-level" thermal specification will be either too tight for robust devices or too loose for sensitive ones.

**Lesson 4: The PI speedup varies by $10^5\times$ across cases.** The speedup ranges from $\sim 10^3\times$ (Cases 1–2, self-energy) to $\sim 10^8\times$ (Case 7, Gutzwiller) to infinite (Case 4, $S = F$). This means the value proposition of the PI approach is *not uniform* — it is most compelling for Cases 4, 5, and 7, where the alternative (Monte Carlo or brute-force diag.) is either impossible or months of compute.

**Lesson 5: Quantum yield advantage is largest at the system level.** Per-component $\Delta Y$ (Cases 1, 2) is modest (a few percentage points). System-level $\Delta Y$ (Case 3 scaling law) grows linearly with component count and can exceed 10 percentage points for $N > 20$. **The business case for quantum operation is a system-level argument, not a component-level one.**

**Lesson 6: Two cases provide qualitatively new physics, not just faster computation.** Cases 6 (PhC, influence functional) and Case 8 (topological) are not merely "faster than MC" — they predict physics that MC on the Hamiltonian *cannot capture at all*: non-Markovian entanglement revival (Case 6) and topological classification of disorder (Case 8). For these cases, the PI is not an optimization — it is the only correct formalism.

**Lesson 7: The binding constraint is always a parity or symmetry property.** Across all 8 cases, the tightest constraint is determined by a symmetry of the disorder: even-parity roughness (Cases 1–4), discrete Maslov index (Case 5), structured-environment spectral shape (Case 6), orbit stability (Case 7), or topological winding number (Case 8). **Symmetry analysis of the disorder — not just its amplitude — is the key to efficient tolerance engineering.**

Table 5: Seven lessons: summary.

| # | Lesson | Practical Implication |
|---|---|---|
| 1 | Tightest tolerance is structural | Check layout topology before tightening fab specs |
| 2 | Continuous, discrete, binary errors differ | Use appropriate statistics for each in combined yield |
| 3 | Thermal is device-class dependent | Allocate thermal budgets per-device, not per-chip |
| 4 | PI speedup varies $10^3$–$10^8\times$ | Prioritize PI for Cases 4, 5, 7 (highest leverage) |
| 5 | $\Delta Y$ grows at system level | Quantum operation business case is system-level |
| 6 | Two cases give new physics, not just speed | Cases 6 and 8: PI is the only correct formalism |
| 7 | Binding constraint is a symmetry property | Analyze disorder symmetry, not just amplitude |

## 6.2 Mixed-Case Systems: When Devices Combine on One Chip

Real photonic integrated circuits are not single devices — they are *systems* that combine multiple device types on the same chip. A typical quantum PIC might include microring SFWM sources

(Case 2), directional coupler beam splitters (Case 1), MZI phase shifters (Case 4), adiabatic taper mode converters (Case 5), and possibly PhC cavity memories (Case 6). The tolerance analysis of such a system requires careful handling of the interactions between cases.

### 6.2.1 How Tolerances Compound in Mixed Systems

**Mixed-System Tolerance Compounding Rules**

1. **Cascaded components (series):** System fidelity is the product of component fidelities: $F_{\text{sys}} = \prod_i F_i$. *The weakest component limits the system.*
2. **Parallel paths (interferometer arms):** Differential errors between paths matter. Common-mode errors cancel (CMRR from Case 1). *Layout symmetry is critical* (Lesson 1).
3. **Resonant × non-resonant:** Ring resonators (Case 2) amplify phase errors by $\sim Q/\pi$ round trips. A coupler (Case 1) upstream of a ring contributes its tolerance *once*; the ring amplifies its *own* tolerance by $Q/\pi$. *The ring is almost always the yield-limiting element in a ring-coupler system.*
4. **Continuous + discrete errors:** The Maslov phase from tapers (Case 5) is *systematic and deterministic*, not statistical. It must be treated as a *bias*, not added to the variance. The combined phase budget is:

$$\phi_{\text{total}} = \underbrace{\phi_{\text{Maslov}}}_{\text{systematic bias}} + \underbrace{\delta\phi_{\text{roughness}}}_{\text{random, Gaussian}} .$$

   Monte Carlo that ignores the bias term will give the wrong mean, not just the wrong variance.
5. **Cross-tolerance correlations:** Thermal expansion causes both index change (Class B) and mechanical stress (Class C) simultaneously. In a mixed system, a temperature excursion can shift a ring off-resonance *and* jitter an MZI fringe *and* shift a taper cutoff. Use the composite $J_{\text{total}}(\omega)$ (Sec. 2) to capture these correlations.

### 6.2.2 Priority Hierarchy: What to Fix First

When the system yield is below specification, the engineer needs a priority hierarchy for deciding which tolerance to tighten first. Based on the eight cases, we propose the following order:

Table 6: Tolerance tightening priority hierarchy for mixed-case PICs.

| Priority | Action | Rationale | Cost |
|---|---|---|---|
| 1 | Fix layout symmetry (Maslov) | Zero cost; eliminates systematic $\pi/2$ errors. Cases 4, 5. | Free (design change only) |
| 2 | Equalize taper counts | Eliminates Maslov mismatch between interferometer arms. Case 5. | Free (layout rerouting) |
| 3 | Switch to $LiNbO_3$ for thermal-critical components | 5× lower thermal sensitivity. Case 4. Eliminates thermal as co-dominant tolerance. | Moderate (platform change) |
| 4 | Tighten even-parity roughness | Targets the binding constraint (40% of disorder variance). Cases 1–4. | High (etch chemistry, resist) |
| 5 | Add thermal stabilization | Removes Class B tolerance for rings. Case 2. | Moderate (heater + PID) |
| 6 | Reduce hole disorder (PhC) | Exponential $Q$ sensitivity. Case 6. | High (e-beam lithography) |
| 7 | Use influence functional (not Lindblad) | Wider tolerance windows for Case 6. | Zero (software change) |
| 8 | Design for quantum operation | $\Delta Y$ improvement from entanglement. System-level. | Moderate (entangled source) |

The hierarchy is ordered by *cost-effectiveness*: free actions first (layout fixes), then moderate-cost actions (platform change, thermal control), then high-cost actions (process improvement), then paradigm changes (quantum operation).

### 6.2.3 Worked Guidance: A Realistic Quantum PIC

Consider a concrete example: a quantum PIC for entangled-pair distribution, consisting of:

- 1 microring SFWM source (Case 2)
- 2 directional coupler beam splitters (Case 1)
- 4 MZI phase shifters (Case 4)
- 8 adiabatic tapers for routing (Case 5)

**Step 1: Identify the yield-limiting element.** The ring (Case 2) has the tightest fabrication tolerance ($\sigma_w < 2$ nm for $Y > 90\%$) and the tightest thermal tolerance ($\Delta T < 1$ K). **The ring is the yield-limiting element.**

**Step 2: Check for Maslov mismatch.** The 8 tapers distribute as 4 per MZI arm. If the layout is symmetric (4 tapers per arm), $\Delta\mu = 0$ and no Maslov alarm. If asymmetric (e.g., 5 vs. 3 due to routing constraints), $\Delta\mu = 2$, giving a systematic $\pi$ rad error per MZI. **This must be fixed at the layout stage.**

**Step 3: Compute system yield.**

- Ring: $S_{\text{ring}} \approx 0.94$ (at $\sigma_w = 2$ nm).
- Couplers: $S_{\text{coupler}} \approx 0.98$ each.
- MZIs: $S_{\text{MZI}} \approx 0.96$ each (differential roughness).
- System (classical): $F_{\text{sys,cl}} = 0.94 \times 0.98^2 \times 0.96^4 \approx 0.72$.

- System (quantum): $F_{\text{sys,qu}} = S_{12,\text{ring}} \times S^2_{12,\text{coup}} \times S^4_{12,\text{MZI}} \approx 0.84$.
- $\Delta Y \approx 12$ percentage points.

**Quantum operation improves system yield from 72% to 84%** — a significant business-case change, and the improvement comes entirely from the entangled light source, with no hardware modification.

**Step 4: Apply the priority hierarchy.**

1. Fix layout symmetry: ensure $\Delta\mu = 0$. (Free.)
2. Add thermal stabilization to the ring: $\Delta T < 1$ K. (Moderate.)
3. If yield still insufficient: tighten even-parity roughness to $\sigma_{w,e} < 0.5$ nm. (High cost.)
4. Operate with entangled source for 12-point $\Delta Y$ improvement. (Moderate: entangled-pair source.)

### 6.2.4 Warning: What Can Go Wrong in Mixed Systems

**Mixed-System Pitfalls**

- **Treating Maslov phase as noise:** The Maslov contribution is a bias, not a variance. Adding it in quadrature (as MC tolerance tools do by default) *underestimates* the phase error. It must be added as a deterministic offset.
- **Using chip-level thermal specs:** A single $\Delta T$ specification for the entire chip is too loose for the ring and too tight for the coupler. Allocate thermal budgets per-device based on Table 5.
- **Ignoring cross-tolerance correlations:** If the ring and MZI share a thermal path (common in compact PICs), a temperature excursion shifts the ring resonance *and* the MZI fringe simultaneously. The composite $J_{\text{total}}(\omega)$ captures this; independent-source analysis does not.
- **Optimizing total roughness instead of even-parity roughness:** Process improvements that reduce total $\sigma_w$ by 20% but shift the parity balance toward even-parity can *worsen* quantum yield. The correct target is even-parity roughness specifically.
- **Forgetting that quantum advantage requires entangled input:** The $\Delta Y$ improvement vanishes for classical or single-photon operation. The source (Case 2 ring SFWM) must be designed and verified to produce entangled pairs with sufficient purity.
- **Assuming the influence functional is always needed:** For devices in the Markovian regime ($\tau_B/\tau_S < 0.1$), Lindblad is adequate and much simpler. Use the bridge #5 alarm (Sec. 7) to decide.

## 7 Bridge Equations: Switching Points, Control, and Alarm Mechanisms

The four bridge equations relevant to tolerance are not merely mathematical identities — they are **operational switching points** where the engineer's strategy must change. Each bridge defines a transition between two regimes (two "pillars"), and crossing the bridge in the wrong direction without awareness leads to either (a) wasted effort (using a more expensive formalism than needed) or (b) incorrect predictions (using a cheaper formalism beyond its validity).

This section defines, for each bridge: *what to monitor*, *what threshold triggers an alarm*, and *what corrective action to take*.

### 7.1 Bridge #2: $S = F$ (Classical Strehl = Quantum Fidelity)

**Bridge Equation**

$S = F = |K_{\text{ab}}(\mathbf{0},\mathbf{0})/K_{\text{id}}(\mathbf{0},\mathbf{0})|^2$. Connects B3 (wave optics) $\leftrightarrow$ B4 (quantum information).

**What to monitor.** Strehl ratio $S$ from the standard Code V / Zemax tolerance analysis. This is computed routinely for every imaging system.

**When the alarm triggers.** The bridge is *always valid* for Gaussian phase errors (the identity is exact). The alarm triggers when the application *changes* from classical to quantum:

- Classical telecom: $S > 0.80$ ($\lambda/14$) is sufficient. No alarm.
- Quantum sensing (NV-diamond): $S > 0.9999$ ($\lambda/2800$) required. **Alarm: the existing classical spec is $45\times$ too loose.**
- Non-Gaussian aberrations (kurtosis $> 2$): the $S = F$ identity breaks down. **Alarm: Maréchal approximation fails; use full PI.**

**Corrective action.** When the quantum-sensing alarm triggers: tighten the wavefront specification by the ratio $\lambda_{\text{quantum}}/\lambda_{\text{classical}}$. For NV-diamond: $(\lambda/14)/(\lambda/2800) = 200\times$ tighter surface quality. When the non-Gaussian alarm triggers: compute fidelity directly from the PI propagator ratio, bypassing the Maréchal exponential.

**Tuning parameter.** Surface quality (polishing, lithography resolution) for free-space optics; sidewall roughness (etch chemistry, resist quality) for integrated photonics. The bridge tells the engineer *exactly how much tighter* the spec must be for quantum operation.

### 7.2 Bridge #4: Gutzwiller ($\rho(E)$ from Periodic Orbits)

**Bridge Equation**

$N(k) = N_{\text{Weyl}}(k) + \sum_p A_p \sin(kL_p - \pi\mu_p/2)/(kL_p)$. Connects B5 (classical trajectories) $\leftrightarrow$ B3 (wave eigenvalues).

**What to monitor.** The stability trace $|\text{Tr}(\mathbf{M}_p)|$ of the shortest periodic orbits, and the Brody parameter $\beta$ (from eigenvalue spacing).

**When the alarm triggers.**

- $\beta < 0.3$ (near-integrable): eigenvalues are Poisson-distributed; the Gutzwiller sum converges slowly. **Alarm: use direct diagonalization, not Gutzwiller.**
- $\beta > 0.7$ (near-chaotic): Gutzwiller is most efficient. No alarm.
- Orbit bifurcation (a stable orbit becomes unstable under deformation): the semiclassical amplitude $A_p$ diverges. **Alarm: uniform approximation needed near bifurcation.**

**Corrective action.** When $\beta < 0.3$: fall back to direct Helmholtz eigenvalue diag. (slower but exact). When orbit bifurcation is detected: switch to uniform semiclassical approximation (Airy-function dressing of the divergent orbit contribution). When $\beta > 0.7$: use Gutzwiller with confidence; the $10^8\times$ speedup applies.

**Tuning parameter.** Deformation $\epsilon$. The engineer *designs* the cavity shape to place it in the regime where Gutzwiller is most efficient and directional emission is optimal. $\beta$ serves as a non-destructive quality metric: measure the spectrum of a fabricated device, fit $\beta$, confirm that the intended deformation was achieved.

### 7.3 Bridge #5: Feynman–Vernon ($\gamma_k = \pi J(\omega_k)$)

**Bridge Equation**

$\gamma_k = \pi J(\omega_k)[1 + \bar{n}_{\text{th}}(\omega_k)]$ (Markovian limit of the influence functional). Connects B5 (environment spectral density) $\leftrightarrow$ B4 (decoherence).

**What to monitor.** The ratio $\tau_B/\tau_S$ (bath correlation time to system decay time). Equivalently: the width of the spectral density $J(\omega)$ relative to the system frequency spacing.

**When the alarm triggers.**

- $\tau_B/\tau_S < 0.1$: safely Markovian. Lindblad is adequate. No alarm.
- $0.1 < \tau_B/\tau_S < 0.3$: marginal. **Alarm: Lindblad may underestimate tolerance window by 10–30%. Verify with IF.**
- $\tau_B/\tau_S > 0.3$: non-Markovian. **Red alarm: Lindblad is qualitatively wrong.** Predicts entanglement death where IF predicts revival. **Must use influence functional.**

**Corrective action.** When the marginal alarm triggers: run both Lindblad and IF; report the discrepancy as a tolerance-window uncertainty. When the red alarm triggers: abandon Lindblad entirely; use the influence functional with the measured or designed $J(\omega)$. The $J(\omega)$ can be *engineered* (by photonic crystal bandgap tuning) to control the non-Markovianity — making the environment a design parameter, not noise.

**Tuning parameter.** The spectral density $J(\omega)$ itself. For PhC cavities: hole size, period, and defect geometry tune the bandgap edge, which controls $J(\omega)$ and hence the non-Markovianity. For thermal environments: thermal resistance $R_{\text{th}}$ and heat capacity $C$ (set by substrate and packaging) determine $\tau_{\text{th}}$.

**Bridge #5 Alarm: When Lindblad Is Dangerous**

The Lindblad equation is the workhorse of open quantum systems in the photonics community. But for structured environments near the strong-coupling/non-Markovian boundary, it can predict **pessimistic** tolerance: telling the engineer that the device fails when it actually works (entanglement revives). This means: (a) good devices rejected by quality control, (b) yield underestimated, (c) design over-specified (tighter tolerances than necessary $\rightarrow$ higher cost). The influence functional gives the correct, wider tolerance window.

### 7.4 Bridge #6: Maslov ($\Delta\phi = -\pi\mu/2$)

**Bridge Equation**

$\Delta\phi_{\text{Maslov}} = -\pi\mu/2, \quad \mu \in \mathbb{Z}_{\geq 0}$. Connects B1 (singularity topology) $\leftrightarrow$ B3 (propagation phase).

**What to monitor.** The number of mode cutoff crossings per routing path in the PIC layout. This can be extracted from the GDS layout file by analyzing the waveguide width profile $w(z)$ along each routing path.

**When the alarm triggers.**

- Symmetric layout ($\Delta\mu = 0$ between interfering paths): no alarm. Maslov phases cancel in differential measurements.
- Asymmetric layout ($\Delta\mu \geq 1$): **Alarm: systematic phase error $\geq \pi/2$ per taper asymmetry.** For an MZI with $\Delta\mu = 1$: $\pi/2$ rad offset, shifting the interference fringe by a quarter wave.
- Routing path with $N_{\text{tapers}} > 10$: **Alarm: accumulated Maslov phase $> 5\pi$, exceeding $2\lambda$ OPD.** This dominates all other phase-error sources.

**Corrective action.** When the asymmetry alarm triggers: **redesign the PIC layout** to equalize the number of tapers in interfering paths. This is a layout constraint, not a fabrication constraint — it costs nothing to fix at the design stage but is catastrophic if missed. When the accumulation alarm triggers: add the Maslov phase budget to the PIC routing specification; verify that downstream interferometers compensate for the accumulated phase.

**Tuning parameter.** PIC layout topology: number and placement of tapers, mode converters, and spot-size converters in each routing path. The engineer's tool is the floor plan, not the fab process.

**Bridge #6 Alarm: The Invisible Phase Error**

The Maslov phase is the **only tolerance source in this catalog that is completely invisible to all commercial photonic EDA tools** (Code V, Lumerical, RSoft, Synopsys). It is systematic (not random), deterministic (not statistical), and accumulates linearly with taper count. A PIC designer using any existing tool will not see it, will not budget for it, and will not understand why the measured fringe visibility is $\pi/2$ off from the simulation. The path integral is the only formalism that tracks it.

### 7.5 Bridge Alarm Summary

Table 7: Bridge alarm summary: monitor, threshold, and corrective action.

| # | Monitor | Alarm Threshold | Action | Tuning Knob |
|---|---|---|---|---|
| 2 | Strehl $S$ | $S < 0.9999$ for quantum | Tighten spec by 45× | Surface / roughness quality |
| 4 | Brody $\beta$ | $\beta < 0.3$ or bifurcation | Switch to diag. or uniform approx. | Deformation $\epsilon$ |
| 5 | $\tau_B/\tau_S$ | $> 0.3$: Lindblad wrong | Use influence functional | $J(\omega)$ via PhC bandgap |
| 6 | $\Delta\mu$ per path | $\Delta\mu \geq 1$: systematic | Equalize taper count | PIC layout topology |

## 8 Implementation: Workflow and Reporting

### 8.1 Companion Code

All figures generated by `self-energy-8cases.py`. Output structure:

```
GAI_SelfEnergy_8Cases_YYYY_MM_DD_HHMM/
  sim_config.json    all_results.json    summary_report.md
```

```
summary_cross_case_comparison.png
case_01_coupler/ ... case_08_mobius/
  (overview plots + highlight.txt each)
```

Update `\FigRoot` in the preamble to match the timestamp.

### 8.2 Quality Checklist

Table 8: Quality checklist for tolerance analysis.

| # | Check Item | Pass? |
|---|---|---|
| 1 | Fabrication disorder PSD obtained | ☐ |
| 2 | Phase-error PSD computed | ☐ |
| 3 | Self-energy $\Delta S$ computed | ☐ |
| 4 | Classical Strehl $S$ computed | ☐ |
| 5 | Biphoton Strehl $S_{12}$ (parity decomposition) | ☐ |
| 6 | Thermal $J_{\mathrm{th}}(\omega)$ constructed | ☐ |
| 7 | Markovian / non-Markovian classified | ☐ |
| 8 | Thermal yield window computed | ☐ |
| 9 | Mechanical PSD characterized (if applicable) | ☐ |
| 10 | Combined phase budget computed | ☐ |
| 11 | $Y_{\mathrm{cl}}$ at threshold | ☐ |
| 12 | $Y_{\mathrm{qu}}$ at threshold | ☐ |
| 13 | $\Delta Y$ reported | ☐ |
| 14 | Dominant tolerance identified | ☐ |
| 15 | Binding constraint $\rightarrow$ design rule | ☐ |
| 16 | Maslov budget included (if tapers) | ☐ |
| 17 | Bridge alarms checked (Table 7) | ☐ |
| 18 | Non-Gaussian defects flagged | ☐ |
| 19 | PI speedup quantified | ☐ |
| 20 | All outputs saved per GAI convention | ☐ |

## 9 The Overarching Finding

**Central Finding**

Tolerance analysis in photonic engineering has been dominated by Monte Carlo simulation for decades. The path-integral approach offers three fundamental advantages:

1. **Speed:** $10^2$–$10^8\times$ faster for the dominant disorder models.
2. **Insight:** Structural features invisible to MC — parity decomposition, Maslov budget, non-Markovian windows, orbit robustness, topological protection.
3. **Quantum yield advantage:** $Y_{\mathrm{qu}} \geq Y_{\mathrm{cl}}$, growing with system complexity.

The bridge equations (Sec. 7) provide operational switching points with explicit alarm thresholds and corrective actions, transforming the PI framework from abstract theory into actionable engineering practice.

The practical implication: **tolerance analysis need not be the bottleneck**. The path integral converts the most time-consuming step in photonic device development from a weeks-long Monte Carlo campaign into an afternoon's analytic calculation — with better physical insight, correct quantum predictions, and an alarm system that tells the engineer when to change strategy.

## References


[1] A. Maréchal, "Étude des effets combinés de la diffraction et des aberrations géométriques sur l'image d'un point lumineux," *Rev. d'Optique* **26**, 257–277 (1947).

[2] M. Born and E. Wolf, *Principles of Optics*, 7th ed. (Cambridge University Press, 1999).

[3] F. P. Payne and J. P. R. Lacey, "A theoretical analysis of scattering loss from planar optical waveguides," *Opt. Quantum Electron.* **26**, 977–986 (1994).

[4] F. Grillot, L. Vivien, S. Laval, D. Pascal, and E. Cassan, "Size influence on the propagation loss induced by sidewall roughness in ultrasmall SOI waveguides," *IEEE Photon. Technol. Lett.* **16**, 1661–1663 (2004).

[5] W. Bogaerts, M. Fiers, and P. Dumon, "Design challenges in silicon photonics," *IEEE J. Sel. Top. Quantum Electron.* **20**, 8202014 (2014).

[6] F. Xia, L. Sekaric, and Y. Vlasov, "Ultracompact optical buffers on a silicon chip," *Nature Photon.* **1**, 65–71 (2007).

[7] E. Engin *et al.*, "Photon pair generation in a silicon micro-ring resonator with reverse bias enhancement," *Opt. Express* **21**, 27826–27834 (2013).

[8] C. K. Hong, Z. Y. Ou, and L. Mandel, "Measurement of subpicosecond time intervals between two photons by interference," *Phys. Rev. Lett.* **59**, 2044–2046 (1987).

[9] A. Politi, M. J. Cryan, J. G. Rarity, S. Yu, and J. L. O'Brien, "Silica-on-silicon waveguide quantum circuits," *Science* **320**, 646–649 (2008).

[10] W. R. Clements, P. C. Humphreys, B. J. Metcalf, W. S. Kolthammer, and I. A. Walmsley, "Optimal design for universal multiport interferometers," *Optica* **3**, 1460–1465 (2016).

[11] D. A. B. Miller, "Self-configuring universal linear optical component," *Photon. Res.* **1**, 1–15 (2013).

[12] J. Carolan *et al.*, "Universal linear optics," *Science* **349**, 711–716 (2015).

[13] S. Aaronson and A. Arkhipov, "The computational complexity of linear optics," *Theory Comput.* **9**, 143–252 (2013).

[14] J. J. Renema *et al.*, "Efficient classical algorithm for boson sampling with partially distinguishable photons," *Phys. Rev. Lett.* **120**, 220502 (2018).

[15] Y. Fu, T. Ye, W. Tang, and T. Chu, "Efficient adiabatic silicon-on-insulator waveguide taper," *Photon. Res.* **2**, A41–A44 (2014).

[16] X. Sun and D. L. Bhatt, "Adiabatic mode transformation in silicon waveguide tapers," *Proc. SPIE* **7220**, 72200G (2009).

[17] V. P. Maslov and M. V. Fedoriuk, *Semi-Classical Approximation in Quantum Mechanics* (Reidel, Dordrecht, 1981).

[18] Y. Akahane, T. Asano, B.-S. Song, and S. Noda, "High-$Q$ photonic nanocavity in a two-dimensional photonic crystal," *Nature* **425**, 944–947 (2003).

[19] Y. Taguchi *et al.*, "Statistical studies of photonic heterostructure nanocavities with an average $Q$ factor of three million," *Opt. Express* **19**, 11916–11921 (2011).

[20] J. P. Reithmaier *et al.*, "Strong coupling in a single quantum dot–semiconductor microcavity system," *Nature* **432**, 197–200 (2004).

[21] T. Yoshie *et al.*, "Vacuum Rabi splitting with a single quantum dot in a photonic crystal nanocavity," *Nature* **432**, 200–203 (2004).

[22] R. P. Feynman and F. L. Vernon, Jr., "The theory of a general quantum system interacting with a linear dissipative system," *Ann. Phys.* **24**, 118–173 (1963).

[23] H.-P. Breuer, E.-M. Laine, and J. Piilo, "Measure for the degree of non-Markovian behavior of quantum processes in open systems," *Phys. Rev. Lett.* **103**, 210401 (2009).

[24] J. U. Nöckel and A. D. Stone, "Ray and wave chaos in asymmetric resonant optical cavities," *Nature* **385**, 45–47 (1997).

[25] C. Gmachl *et al.*, "High-power directional emission from microlasers with chaotic resonators," *Science* **280**, 1556–1564 (1998).

[26] M. C. Gutzwiller, *Chaos in Classical and Quantum Mechanics* (Springer, New York, 1990).

[27] E. B. Bogomolny, "Semiclassical quantization of multidimensional systems," *Nonlinearity* **5**, 805–866 (1992).

[28] S. C. Creagh and R. G. Whelan, "Homoclinic structure controls the optical properties of deformed dielectric microcavities," *Phys. Rev. Lett.* **82**, 5237–5240 (1999).

[29] M. Hafezi, S. Mittal, J. Fan, A. Migdall, and J. M. Taylor, "Imaging topological edge states in silicon photonics," *Nature Photon.* **7**, 1001–1005 (2013).

[30] M. C. Rechtsman *et al.*, "Photonic Floquet topological insulators," *Nature* **496**, 196–200 (2013).

[31] K. Y. Bliokh, F. J. Rodríguez-Fortuño, F. Nori, and A. V. Zayats, "Spin-orbit interactions of light," *Nature Photon.* **9**, 796–808 (2015).

[32] R. Chen, F. Yu, J. Wang, J. Chen, R. Jin, J. Wang, G. Li, X. Chen, and W. Lu, "Möbius metasurface for fully decoupled bidirectional light control," *Adv. Photon.* **8**(2), 026005 (2026).

[33] J. P. Dowling, "Quantum optical metrology — the lowdown on high-N00N states," *Contemp. Phys.* **49**, 125–143 (2008).

[34] C. M. Caves, "Quantum-mechanical noise in an interferometer," *Phys. Rev. D* **23**, 1693–1708 (1981).